

Opinion Perception Survey Analysis (preregistration)

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Invalid Date

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```
load_or_install <- function(pkg) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  }
}

pkgs <- c(
  "tidyverse", "lme4", "lmerTest", "nloptr", "httpgd",
  "languageserver", "lmtest", "sandwich", "caret", "knitr",
  "prettycode", "performance", "estimatr", "patchwork", "broom"
)

library("broom.mixed")
knitr::opts_chunk$set(comment = NA)
invisible(lapply(pkgs, load_or_install))
options(vsc.viewer = TRUE)
```

```
theme(
  text = element_text(family = "Noto Sans")
)
options(tibble.print_max = Inf)
options(tibble.width = Inf)
```

1 Hypotheses and analysis plan

All hypotheses concerning a single slope parameter (e.g., H1 to H3) are tested using a t-test at significance level $\alpha = 0.05$. Hypotheses involving multiple parameters are tested using a single F-test on their joint significance, also at $\alpha = 0.05$. All subsequent hypotheses follow this convention without further repetition.

```
# -- Standardised test-result printers -----

#' One-sided t-test result block (for single beta)
print_one_sided <- function(label, beta, se, t, df = Inf,
  direction = c("negative", "positive"),
  ci = NULL) {
  direction <- match.arg(direction)
  p_two <- 2 * pt(-abs(t), df = df)
  p_one <- if ((direction == "negative" && beta < 0) ||
    (direction == "positive" && beta > 0)) p_two / 2 else 1 - p_two / 2
  supported <- p_one < 0.05 && ((direction == "negative" && beta < 0) ||
    (direction == "positive" && beta > 0))

  cat(sprintf("=== %s ===\n", label))
  cat(sprintf(" Estimate = %8.4f\n", beta))
  cat(sprintf(" SE = %8.4f\n", se))
  cat(sprintf(" t = %8.4f\n", t))
  cat(sprintf(" df = %s\n", ifelse(is.finite(df), sprintf("%.1f", df), "infy")))
  if (!is.null(ci))
    cat(sprintf(" 95%% CI = [%.4f, %.4f]\n", ci[1], ci[2]))
  cat(sprintf(" p (one-sided, beta %s 0) = %.4f\n",
    ifelse(direction == "negative", "<", ">"), p_one))
  cat(sprintf(" Supported : %s\n\n", ifelse(supported, "YES", "NO")))
  invisible(list(beta = beta, t = t, p_one = p_one, supported = supported))
}

#' LRT / F-test result block
print_lrt <- function(label, stat, df_num, df_den = NULL, p,
  type = c("chi2", "F")) {
  type <- match.arg(type)
  cat(sprintf("=== %s ===\n", label))
  if (type == "chi2")
    cat(sprintf(" chi^2(%d) = %.3f, p = %.4f %s\n\n",
      df_num, stat, p, sig_stars(p)))
  else
    cat(sprintf(" F(%d, %d) = %.3f, p = %.4f %s\n\n",
      df_num, df_den, stat, p, sig_stars(p)))
  invisible(list(stat = stat, p = p))
}

#' Model-fit comparison table
print_fit_table <- function(...) {
  models <- list(...)
  tbl <- data.frame(
    Model = names(models),
    AIC = round(sapply(models, AIC), 1),
    BIC = round(sapply(models, BIC), 1),
    logLik = round(sapply(models, function(m) as.numeric(logLik(m))), 1),
    npar = sapply(models, function(m) length(fixef(m)))
  )
  tbl$delta_AIC <- tbl$AIC - min(tbl$AIC)
```

```
tbl$delta_BIC <- tbl$BIC - min(tbl$BIC)
print(tbl, row.names = FALSE)
invisible(tbl)
}

tidy_print <- function(m, mixed = FALSE) {
  tidy_fn <- if (mixed) broom.mixed::tidy else broom::tidy
  tidy_fn(m, effects = "fixed") |>
    select(term, estimate, std.error, p.value) |>
    mutate(across(where(is.numeric), \(x) round(x, 3))) |>
    print(m=Inf) |>
    knitr::kable()
}

sig_stars <- function(p) {
  case_when(p < .001 ~ "***", p < .01 ~ "**", p < .05 ~ "*",
            p < .1 ~ ".", TRUE ~ "ns")
}

# Extract lmer beta, SE, t, Satterthwaite df in one call
lmer_coef <- function(model, term) {
  cf <- summary(model)$coefficients
  list(
    beta = cf[term, "Estimate"],
    se = cf[term, "Std. Error"],
    t = cf[term, "t value"],
    df = cf[term, "df"], # added by lmerTest
    p2 = cf[term, "Pr(>|t|)"]
  )
}
```

```
questions_sc <- c(
  "climate_concern", "gay_marriage", "rights_indep_integration",
  "econ_inequality", "regulate_internet", "east_germans"
)

parties <- c("Left Party", "BSW", "Green Party", "SPD", "FDP", "CDU/CSU",
             "AfD", "No party", "Other party", "Refuse to say/No answer")
partiesVars <- c("LeftParty", "BSW", "GreenParty", "SPD", "FDP", "CDU/CSU", "AfD")
partiesCloseSel <- c("Left Party", "BSW", "Green Party", "SPD", "FDP", "CDU/CSU", "AfD")

# Shared column-name vectors (used across many hypotheses)
col_deltas <- paste0("deltaX_", questions_sc)
col_alpha_exp <- paste0("exp_alpha_", questions_sc)
col_alpha_lin <- paste0("linear_alpha_", questions_sc)
col_weights <- paste0("w_", questions_sc)
col_var_std <- paste0("std_socialCircle_ops_", questions_sc)
col_var_var <- paste0("var_socialCircle_ops_", questions_sc)
```

2 Load Data

```
df_p <- read.csv(
  "processed_data/2026-06-19_data_processed_participant_withAllIssueWeights.csv"
)

inds_bothwaves <- df_p |>
  count(id) |>
  filter(n == 2) |>
  pull(id)

treatment_ids <- df_p |>
  filter(wave == 2, treatment_wave2 == TRUE, id %in% inds_bothwaves) |>
  pull(id)
```

```

control_ids <- df_p |>
  filter(wave == 2, treatment_wave2 == FALSE, id %in% inds_bothwaves) |>
  pull(id)

df_p <- df_p |>
  mutate(
    overall_std_socialCircle = rowMeans(across(all_of(col_var_std))),
    across(all_of(col_var_std), function(x) x^2,
      .names = "var_{gsub('std_', '', col)}"),
    wave = factor(wave)
  )

df_diff <- read.csv(
  "processed_data/2026-06-19_data_processed_differences_withAllIssueWeights.csv"
) |>
  mutate(
    ingroupdummy = as.integer(ingroupdummy == "True"),
    wave = factor(wave)
  ) |>
  left_join(
    df_p |> select(id, wave,
      all_of(paste0("var_socialCircle_ops_", questions_sc)),
      all_of(paste0("std_socialCircle_ops_", questions_sc))),
    by = c("id", "wave")
  )

```

3 Hypotheses

3.1 Aim 1

Aim 1: Compare approaches to measuring perceived political differences

3.1.1 H1

3.1.1.1 H1. Participants place the typical voter of the party they affiliate with nearer to themselves than other typical voters.

We model the distance between the dots representing the participant (self) and the typical voter of each party r as

$$d_{n,\text{self},r} = \beta_0 + \beta_1 \mathbb{I}(r = \text{party}_n) + u_n + \epsilon_{n,r}, \quad (1)$$

where $\mathbb{I}(\cdot)$ is an indicator function, u_n is a participant level random intercept, and $\epsilon_{n,r}$ is an error term.

The prediction is $\beta_1 < 0$. Participants who prefer a political party place the typical voter of that party closer to themselves than the average of the typical voters of other parties.

```

df_h1 <- df_diff |>
  filter(wave %in% c(1, 2), dot1 == "self", dot2 %in% partiesVars) |>
  select(id, wave, ingroupdummy, party, pixel_dist)

m_h1 <- lmer(pixel_dist ~ ingroupdummy + wave + (1 | id), data = df_h1, REML = TRUE)
ci_h1 <- confint(m_h1, parm = "ingroupdummy", method = "Wald")
cf_h1 <- lmer_coef(m_h1, "ingroupdummy")

print_one_sided("H1: Ingroup party placed closer to self",
  cf_h1$beta, cf_h1$se, cf_h1$t, cf_h1$df,
  direction = "negative", ci = ci_h1)

```

=== H1: Ingroup party placed closer to self ===

```

Estimate = -0.2520
SE       =  0.0041
t        = -61.5081
df       = 13785.6
95% CI   = [-0.2600, -0.2440]
p (one-sided, beta < 0) = 0.0000
Supported : YES

```

3.1.2 H2

3.1.2.1 H2. Distances to voters on the map correspond to likeability ratings.

We model likeability ratings of the typical voter of political party r (rated only in wave 2) as a function of map distances:

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \quad (2)$$

The prediction is $\beta_1 < 0$. Voters placed farther away from the participant's own dot on the map receive lower likeability ratings than voters placed closer to the participant's own dot.

```
df_h2 <- df_diff |>
  filter(wave == 2, dot1 == "self", dot2 %in% partiesVars) |>
  select(id, sympathy, party, pixel_dist)

m_h2 <- lmer(sympathy ~ pixel_dist + (1 | id), data = df_h2, REML = TRUE)
ci_h2 <- confint(m_h2, parm = "pixel_dist", method = "Wald")
cf_h2 <- lmer_coef(m_h2, "pixel_dist")

print_one_sided("H2: Map distance predicts likeability",
  cf_h2$beta, cf_h2$se, cf_h2$t, cf_h2$df,
  direction = "negative", ci = ci_h2)
```

```
=== H2: Map distance predicts likeability ===
Estimate = -1.1108
SE       = 0.0160
t        = -69.6313
df       = 6723.9
95% CI   = [-1.1421, -1.0795]
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

3.1.3 H3

3.1.3.1 H3. Distances on the political map correspond to similarity ratings.

Similar to H2, we model similarity ratings between a pair of individuals, i and j , as a function of map distances:

$$s_{n,ij} = \beta_0 + \beta_1 \cdot d_{n,ij} + u_n + \epsilon_{n,ij}. \quad (3)$$

The prediction is $\beta_1 < 0$. Pairs placed farther apart on the map receive lower similarity ratings than pairs placed closer together.

```
df_h3 <- df_diff |>
  filter(wave %in% c(1, 2)) |>
  select(id, wave, pairwise_similarity, pixel_dist) |>
  drop_na()

m_h3 <- lmer(pairwise_similarity ~ pixel_dist + wave + (1 | id), data = df_h3, REML = TRUE)
ci_h3 <- confint(m_h3, parm = "pixel_dist", method = "Wald")
cf_h3 <- lmer_coef(m_h3, "pixel_dist")

print_one_sided("H3: Map distance predicts pairwise similarity",
  cf_h3$beta, cf_h3$se, cf_h3$t, cf_h3$df,
  direction = "negative", ci = ci_h3)
```

```
=== H3: Map distance predicts pairwise similarity ===
Estimate = -0.7761
SE       = 0.0066
t        = -117.4882
df       = 38233.2
95% CI   = [-0.7890, -0.7631]
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

3.1.4 H4

3.1.4.1 H4. The closest pair on the map is rated most similar and the farthest pair is rated least similar

Consider $(ij)^{min}$ the pair of individuals i and j with the smallest distance on participant n 's political map and $(ij)^{max}$ the pair with the largest distance on the map. The similarity difference between these two pairs

$$\Delta s_{max} = s_{n,(ij)^{min}} - s_{n,(ij)^{max}} . \quad (4)$$

The prediction is $\Delta s_{max} > 0$ (standard t-test with $\alpha = 0.05$). The closest pair on the map receives a higher similarity rating than the farthest pair.

```
df_h4_wide <- df_diff |>
  filter(wave %in% c(1, 2),
         minPixelDistance_pair_dummy == "True" | maxPixelDistance_pair_dummy == "True") |>
  select(id, wave, maxPixelDistance_pair_dummy, pairwise_similarity) |>
  pivot_wider(
    id_cols = c(id, wave),
    names_from = maxPixelDistance_pair_dummy,
    values_from = pairwise_similarity
  ) |>
  rename(sim_min = "False", sim_max = "True") |>
  mutate(diff = sim_min - sim_max)

t_h4 <- t.test(df_h4_wide$diff, mu = 0)

cat("=== H4: Closest pair rated most similar ===\n")
cat(sprintf(" Mean(sim_min) = %.4f ± %.4f\n", mean(df_h4_wide$sim_min), sd(df_h4_wide$sim_min)))
cat(sprintf(" Mean(sim_max) = %.4f ± %.4f\n", mean(df_h4_wide$sim_max), sd(df_h4_wide$sim_max)))
cat(sprintf(" Mean diff = %.4f\n", mean(df_h4_wide$diff)))
cat(sprintf(" t = %.4f\n", t_h4$statistic))
cat(sprintf(" p (one-sided) = %.4f\n", t_h4$p.value / 2))
cat(sprintf(" Supported : %s\n",
            ifelse(t_h4$statistic > 0 && t_h4$p.value / 2 < 0.05, "YES", "NO")))
```

```
=== H4: Closest pair rated most similar ===
Mean(sim_min) = 0.7113 ± 0.2688
Mean(sim_max) = 0.2308 ± 0.2771
Mean diff = 0.4805
t = 55.3777
p (one-sided) = 0.0000
Supported : YES
```

3.1.5 H5

3.1.5.1 H5. The inferred issue weights are robust across waves.

We test the reliability of our tool by comparing the similarity of issue weights, $\alpha_{n,q}$, of participants in wave 1 vs.~wave 2 (control condition) with the similarity of issue weights between participants.

$$\overline{\Delta\alpha}_{q_{\text{within}}} = \mathbb{E} [|\alpha_{n,q}(\text{wave 2}) - \alpha_{n,q}(\text{wave 1})|] \quad \text{and} \quad (5)$$

$$\overline{\Delta\alpha}_{q_{\text{between}}} = \mathbb{E} [|\alpha_{n,q}(\text{wave 2}) - \alpha_{m,q}(\text{wave 1/wave 2})|] \quad \text{with } m \neq n \quad (6)$$

ERROR: It should be: $\overline{\Delta\alpha}_{q_{\text{within}}} - \overline{\Delta\alpha}_{q_{\text{between}}} < 0$

ADDON: $\mathbb{E}$ is the L1 (Manhattan) norm, i.e. $\mathbb{E} [|\alpha_{n,q}(\text{wave 2}) - \alpha_{m,q}(\text{wave 1})|] = \sum_q [|\alpha_{n,q}(\text{wave 2}) - \alpha_{m,q}(\text{wave 1})|]$.

ADDON: I compare only wave 2 alphas of participant n to wave 1 alphas of all other participants m .

The prediction is that $\overline{\Delta\alpha}_{q_{\text{within}}} - \overline{\Delta\alpha}_{q_{\text{between}}} < 0$ (note: corrected error from preregistration) (one-sample t-test with $\alpha = 0.05$). Participants' issue weights in wave 2 are more similar to their own issue weights in the previous wave than to those of other participants in wave 1 (in the control condition).

```
k <- "exp"
alpha_cols <- paste0(k, "_alpha_", questions_sc)

ctrl_ids_h5 <- df_p |>
  filter(id %in% inds_bothwaves, wave == 2, treatment_wave2 == "0") |>
  pull(id)

df_h5 <- df_p |>
```

```

filter(if_all(all_of(alpha_cols), Negate(is.na)), id %in% ctrl_ids_h5) |>
select(id, wave, all_of(alpha_cols))

ids_both <- intersect(
  df_h5 |> filter(wave == 1) |> pull(id),
  df_h5 |> filter(wave == 2) |> pull(id)
)
df_h5 <- df_h5 |> filter(id %in% ids_both)

w1_mat <- df_h5 |> filter(wave == 1) |> arrange(id) |> select(all_of(alpha_cols)) |> as.matrix()
w2_mat <- df_h5 |> filter(wave == 2) |> arrange(id) |> select(all_of(alpha_cols)) |> as.matrix()
n      <- nrow(w1_mat)

within_means <- rowMeans(abs(w2_mat - w1_mat))
between_means <- vapply(seq_len(n), function(i) {
  mean(rowMeans(abs(sweep(w2_mat[-i, , drop = FALSE], 2, w1_mat[i, ]))))
}, numeric(1))

delta      <- within_means - between_means
t_h5      <- t.test(delta, mu = 0, alternative = "two.sided")
cohens_d <- mean(delta) / sd(delta)

cat("=== H5: Issue weights reliable across waves ===\n")
cat(sprintf("  Within (mean ± sd) = %.4f ± %.4f\n", mean(within_means), sd(within_means)))
cat(sprintf("  Between (mean ± sd) = %.4f ± %.4f\n", mean(between_means), sd(between_means)))
cat(sprintf("  Delta (mean) = %.4f\n", mean(delta)))
cat(sprintf("  Cohen's d = %.4f\n", cohens_d))
cat(sprintf("  t = %.4f\n", t_h5$statistic))
cat(sprintf("  p (two-sided) = %.4f\n", t_h5$p.value))
cat(sprintf("  p (one-sided) = %.4f\n", t_h5$p.value / 2))
cat(sprintf("  Supported : %s\n\n",
  ifelse(t_h5$statistic < 0 && t_h5$p.value / 2 < 0.05, "YES", "NO")))

=== H5: Issue weights reliable across waves ===
  Within (mean ± sd) = 0.1988 ± 0.0777
  Between (mean ± sd) = 0.2211 ± 0.0259
  Delta (mean) = -0.0223
  Cohen's d = -0.3094
  t = -6.1494
  p (two-sided) = 0.0000
  p (one-sided) = 0.0000
  Supported : YES

```

3.1.6 Exploration

3.1.6.1 Explorative Analyses.

In addition to the preregistered hypotheses above, we will explore how people rate the political mapping task against the pairwise similarity task in terms of difficulty and enjoyment, how satisfied they are with the political map, and how much time they needed for each task.

(see 03_data-overview)

3.2 Aim 2

Aim 2: Identify patterns linked to (political) identity

In this section, we hypothesise that demographic and identity-related variables, including party preference, $party_n$, left-right placement, lr_n , age, age_n , or combinations of them moderate the relationship between belief differences and issue weights.

We have two different interpretations of the following analysis: We can analyse the effect per issue separately (eq 2) or jointly across issues (eq 1).

Equation for Interpretation (1) for example for lr :

$$\alpha_{n,q} = \beta_{\alpha_0} + \beta_{lr} \cdot lr + \sum_q \beta_q \cdot \mathbb{I}[\text{issue} = q] + \sum_q \beta_{lr \times q} lr \cdot \mathbb{I}[\text{issue} = q] + (u_n) + \beta_w \text{wave} + \epsilon_{n,q} \quad (7)$$

and equation for interpretation (2) is for each issue q :

$$\alpha_{n,q} = \beta_{\alpha_0} + \beta_{lr} \cdot lr + (u_n) + \beta_w \text{wave} + \epsilon_{n,q} \quad (8)$$

where u_n is the participant-level random intercept (gaussian distributed), included for weights defined as correlations (via lmer) but dropped for weights defined as the fitted weights with linear/exp kernels due to singular fits, where clustered standard errors are used instead

```
run_H_joint <- function(k, predictor, label, use_lmer = TRUE, categorical = FALSE) {
  alpha_cols <- paste0(k, "_alpha_", questions_sc)

  df_h <- df_p |>
    filter(wave %in% c(1, 2)) |>
    select(id, wave, all_of(predictor), all_of(alpha_cols)) |>
    pivot_longer(
      cols = all_of(alpha_cols),
      names_to = "issue",
      names_prefix = paste0(k, "_alpha_"),
      values_to = "alpha"
    ) |>
    filter(!is.na(.data[[predictor]]), !is.na(alpha)) |>
    mutate(
      issue = relevel(factor(issue), ref = "regulate_internet"),
      wave = factor(wave)
    )

  if (categorical) {
    df_h <- df_h |>
      mutate(!predictor := relevel(factor(.data[[predictor]]), ref = "No party"))
  }

  f_null <- alpha ~ issue + wave
  f_full <- reformulate(c("issue", paste0(predictor, " * issue"), "wave"), response = "alpha")

  if (use_lmer) {
    m_null <- lmer(update(f_null, . ~ . + (1 | id)), data = df_h, REML=FALSE)
    m_full <- lmer(update(f_full, . ~ . + (1 | id)), data = df_h, REML=FALSE)
    wt <- anova(m_null, m_full)
    stat <- wt$Chisq[2]; df_num <- wt$Df[2]; p <- wt$`Pr(>Chisq)`[2]
    print_fit_table("Null" = m_null, "Full" = m_full)
    tidy_print(m_full, mixed = TRUE)
    print_lrt(sprintf("%s (issue weights: %s): %s × issue interaction", label, k, predictor),
      stat, df_num, p = p, type = "chi2")
  } else {
    m_null <- lm(f_null, data = df_h)
    m_full <- lm(f_full, data = df_h)
    wt <- anova(m_null, m_full)
    stat <- wt$F[2]; df_num <- wt$Df[2]; df_den <- wt$Res.Df[2]; p <- wt$`Pr(>F)`[2]
    m_full_rob <- lm_robust(f_full, data = df_h, clusters = id, se_type = "CR2")
    cat(sprintf("\nR^2 Null = %.3f | Full = %.3f\n",
      summary(m_null)$r.squared, summary(m_full)$r.squared))
    tidy_print(m_full_rob)
    print_lrt(sprintf("%s (issue weights: %s): %s × issue interaction", label, k, predictor),
      stat, df_num, df_den, p, type = "F")
  }
}
```

```
run_H_per_issue <- function(k, predictor, label, use_lmer = TRUE, categorical = FALSE) {
  cat(sprintf("\n-- %s per issue (issue weight: %s) --", label, k))
```



```

purrr::walk(questions_sc, function(q) {
  df_q <- df_p |>
  filter(wave %in% c(1, 2)) |>
  select(id, wave, all_of(predictor), alpha_q = all_of(paste0(k, "_alpha_", q))) |>
  filter(!is.na(.data[[predictor]]), !is.na(alpha_q)) |>
  mutate(wave = factor(wave))

  if (categorical) {
    df_q <- df_q |>
    mutate(!predictor := relevel(factor(.data[[predictor]]), ref = "No party"))
  }

  f <- reformulate(c(predictor, "wave"), response = "alpha_q")
  f_null <- alpha_q ~ wave

  if (use_lmer) {
    m <- lmer(update(f, . ~ . + (1 | id)), data = df_q, REML = FALSE)
    m_null <- lmer(update(f_null, . ~ . + (1 | id)), data = df_q, REML = FALSE)
    wt <- anova(m_null, m)
    p <- wt$`Pr(>Chisq)`[2]; chi <- wt$Chisq[2]; df_lrt <- wt$Df[2]
    r2v <- r2(m)
    cat(sprintf("\n%s R^2=%.3f/%.3f LRT chi^2(%d)=%.2f p=%.4f %s ",
      q, r2v$R2_marginal, r2v$R2_conditional,
      df_lrt, chi, p, sig_stars(p)))
  } else {
    m <- lm_robust(f, data = df_q, clusters = id, se_type = "CR2")
    m_lm <- lm(f, data = df_q)
    m_null <- lm(f_null, data = df_q)
    wt <- anova(m_null, m_lm)
    p <- wt$`Pr(>F)`[2]; f_stat <- wt$F[2]
    cat(sprintf("\n%s R^2=%.3f F(%d,%d)=%.2f p=%.4f %s ",
      q, summary(m_lm)$r.squared,
      wt$Df[2], wt$Res.Df[2], f_stat, p, sig_stars(p)))
  }
})
}

```

3.2.1 H6

3.2.1.1 H6. Party preference predicts issue weights.

We model the (normalised) issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \sum_r \beta_{r,q} \cdot \mathbb{I}[\text{party}_n = r] + u_q + \epsilon_{n,q} \quad (9)$$

where $\mathbb{I}[\text{party}_n = r]$ are dummy variables for each party r (with one party as reference category), $\beta_{r,q}$ is the effect of party r on weight assigned to issue q , u_q is the average deviation of the weight for issue q , and $\epsilon_{n,q}$ is the individual- and issue-specific error.

The prediction is $\beta_{r,q} \neq 0$ and $\beta_{r1,q} \neq \beta_{r2,q}$ for at least some parties $r1$ and $r2$. Participants affiliated with different parties differ in how strongly they weigh different political issues when mapping individuals.

Different interpretations: Per Issue (1) or Jointly across issues (2)?

Equation for Interpretation (1):

$$\alpha_{n,q} = \beta_0 + \sum_r \beta_r \cdot \mathbb{I}[\text{party}_n = r] + \sum_q \beta_q q + (u_n) + v_q + \epsilon_{n,q} \quad (10)$$

and equation for interpretation (2) is for each issue q :

$$\alpha_{n,q} = \beta_0 + \sum_r \beta_r \cdot \mathbb{I}[\text{party}_n = r] + (u_n) + v_q + \epsilon_{n,q} \quad (11)$$

where u_n is removed if the alphas are constrained (in the case of linear/exp fit).

Interpretation 1: Do parties differ across issues?

For exponential/linear weights (constrained):

```
# H6 - party
run_H_joint ("linear", "party_close", "H6", use_lmer = FALSE, categorical = TRUE)
run_H_joint ("corrP", "party_close", "H6", use_lmer = TRUE, categorical = TRUE)
```

R² Null = 0.009 | Full = 0.016

	term	estimate
1	(Intercept)	0.142
2	issueclimate_concern	0.044
3	issueeast_germans	0.022
4	issueecon_inequality	0.031
5	issuegay_marriage	-0.004
6	issuerights_indep_integration	0.055
7	party_closeAfD	-0.016
8	party_closeBSW	-0.051
9	party_closeCDU/CSU	-0.001
10	party_closeFDP	-0.004
11	party_closeGreen Party	-0.026
12	party_closeLeft Party	0.001
13	party_closeOther party	-0.022
14	party_closeRefuse to say/No answer	0.055
15	party_closeSPD	0.013
16	wave2	0.000
17	issueclimate_concern:party_closeAfD	0.025
18	issueeast_germans:party_closeAfD	0.009
19	issueecon_inequality:party_closeAfD	0.012
20	issuegay_marriage:party_closeAfD	0.014
21	issuerights_indep_integration:party_closeAfD	0.034
22	issueclimate_concern:party_closeBSW	0.082
23	issueeast_germans:party_closeBSW	0.082
24	issueecon_inequality:party_closeBSW	0.063
25	issuegay_marriage:party_closeBSW	0.056
26	issuerights_indep_integration:party_closeBSW	0.023
27	issueclimate_concern:party_closeCDU/CSU	-0.013
28	issueeast_germans:party_closeCDU/CSU	0.022
29	issueecon_inequality:party_closeCDU/CSU	-0.009
30	issuegay_marriage:party_closeCDU/CSU	0.025
31	issuerights_indep_integration:party_closeCDU/CSU	-0.017
32	issueclimate_concern:party_closeFDP	-0.024
33	issueeast_germans:party_closeFDP	0.004
34	issueecon_inequality:party_closeFDP	-0.006
35	issuegay_marriage:party_closeFDP	0.017
36	issuerights_indep_integration:party_closeFDP	0.031
37	issueclimate_concern:party_closeGreen Party	0.060
38	issueeast_germans:party_closeGreen Party	0.017
39	issueecon_inequality:party_closeGreen Party	0.002
40	issuegay_marriage:party_closeGreen Party	0.059
41	issuerights_indep_integration:party_closeGreen Party	0.017
42	issueclimate_concern:party_closeLeft Party	0.056
43	issueeast_germans:party_closeLeft Party	-0.070
44	issueecon_inequality:party_closeLeft Party	-0.027
45	issuegay_marriage:party_closeLeft Party	0.010
46	issuerights_indep_integration:party_closeLeft Party	0.022
47	issueclimate_concern:party_closeOther party	0.045
48	issueeast_germans:party_closeOther party	0.022
49	issueecon_inequality:party_closeOther party	0.006
50	issuegay_marriage:party_closeOther party	0.057
51	issuerights_indep_integration:party_closeOther party	0.001
52	issueclimate_concern:party_closeRefuse to say/No answer	-0.061
53	issueeast_germans:party_closeRefuse to say/No answer	-0.042
54	issueecon_inequality:party_closeRefuse to say/No answer	-0.053
55	issuegay_marriage:party_closeRefuse to say/No answer	-0.033
56	issuerights_indep_integration:party_closeRefuse to say/No answer	-0.144

57		issueclimate_concern:party_closeSPD	-0.007
58		issueeast_germans:party_closeSPD	-0.013
59		issueecon_inequality:party_closeSPD	-0.025
60		issuegay_marriage:party_closeSPD	0.007
61		issuerights_indep_integration:party_closeSPD	-0.042
	std.error	p.value	
1	0.012	0.000	
2	0.019	0.024	
3	0.019	0.252	
4	0.019	0.103	
5	0.017	0.798	
6	0.018	0.002	
7	0.016	0.324	
8	0.021	0.017	
9	0.015	0.934	
10	0.024	0.871	
11	0.015	0.094	
12	0.021	0.943	
13	0.026	0.410	
14	0.060	0.368	
15	0.017	0.433	
16	0.000	0.811	
17	0.027	0.360	
18	0.027	0.738	
19	0.026	0.634	
20	0.025	0.581	
21	0.026	0.191	
22	0.045	0.075	
23	0.043	0.058	
24	0.039	0.109	
25	0.035	0.119	
26	0.038	0.543	
27	0.025	0.594	
28	0.025	0.384	
29	0.024	0.700	
30	0.023	0.289	
31	0.024	0.483	
32	0.034	0.492	
33	0.041	0.919	
34	0.039	0.884	
35	0.035	0.619	
36	0.047	0.506	
37	0.027	0.029	
38	0.026	0.520	
39	0.025	0.925	
40	0.025	0.019	
41	0.025	0.509	
42	0.034	0.103	
43	0.029	0.019	
44	0.032	0.407	
45	0.031	0.748	
46	0.035	0.538	
47	0.048	0.349	
48	0.056	0.694	
49	0.044	0.889	
50	0.046	0.217	
51	0.045	0.978	
52	0.093	0.523	
53	0.089	0.644	
54	0.082	0.525	
55	0.073	0.662	
56	0.081	0.093	
57	0.028	0.795	
58	0.027	0.628	

```
59      0.026  0.353
60      0.026  0.789
61      0.026  0.104
```

```
=== H6 (issue weights: linear): party_close * issue interaction ===
      F(54, 12713) = 1.553,  p = 0.0058  **
```

```
Model      AIC      BIC logLik npar delta_AIC delta_BIC
Null -2358.8 -2291.7 1188.4   7      284.9      0.0
Full -2643.7 -2174.1 1384.9  61        0.0     117.6
```

```
# A tibble: 61 x 4
```

term	estimate
<chr>	<dbl>
1 (Intercept)	0.201
2 issueclimate_concern	0.094
3 issueeast_germans	0.018
4 issueecon_inequality	0.037
5 issuegay_marriage	0.06
6 issuerights_indep_integration	0.114
7 party_closeAfD	0.04
8 party_closeBSW	-0.022
9 party_closeCDU/CSU	-0.039
10 party_closeFDP	-0.06
11 party_closeGreen Party	0.044
12 party_closeLeft Party	-0.003
13 party_closeOther party	-0.084
14 party_closeRefuse to say/No answer	0.138
15 party_closeSPD	-0.007
16 wave2	0.006
17 issueclimate_concern:party_closeAfD	0.011
18 issueeast_germans:party_closeAfD	-0.011
19 issueecon_inequality:party_closeAfD	-0.014
20 issuegay_marriage:party_closeAfD	-0.055
21 issuerights_indep_integration:party_closeAfD	0.027
22 issueclimate_concern:party_closeBSW	0.042
23 issueeast_germans:party_closeBSW	0.023
24 issueecon_inequality:party_closeBSW	0.027
25 issuegay_marriage:party_closeBSW	0.02
26 issuerights_indep_integration:party_closeBSW	-0.006
27 issueclimate_concern:party_closeCDU/CSU	-0.023
28 issueeast_germans:party_closeCDU/CSU	0.029
29 issueecon_inequality:party_closeCDU/CSU	-0.01
30 issuegay_marriage:party_closeCDU/CSU	0.007
31 issuerights_indep_integration:party_closeCDU/CSU	-0.01
32 issueclimate_concern:party_closeFDP	0.035
33 issueeast_germans:party_closeFDP	0.084
34 issueecon_inequality:party_closeFDP	0.041
35 issuegay_marriage:party_closeFDP	0.047
36 issuerights_indep_integration:party_closeFDP	0.077
37 issueclimate_concern:party_closeGreen Party	0.092
38 issueeast_germans:party_closeGreen Party	-0.018
39 issueecon_inequality:party_closeGreen Party	0.026
40 issuegay_marriage:party_closeGreen Party	0.117
41 issuerights_indep_integration:party_closeGreen Party	0.037
42 issueclimate_concern:party_closeLeft Party	0.137
43 issueeast_germans:party_closeLeft Party	-0.036
44 issueecon_inequality:party_closeLeft Party	0.109
45 issuegay_marriage:party_closeLeft Party	0.149
46 issuerights_indep_integration:party_closeLeft Party	0.095
47 issueclimate_concern:party_closeOther party	0.062
48 issueeast_germans:party_closeOther party	0.004
49 issueecon_inequality:party_closeOther party	0.048
50 issuegay_marriage:party_closeOther party	0.041
51 issuerights_indep_integration:party_closeOther party	0.025
52 issueclimate_concern:party_closeRefuse to say/No answer	-0.112

53	issueeast_germans:party_closeRefuse to say/No answer	-0.076
54	issueecon_inequality:party_closeRefuse to say/No answer	-0.035
55	issuegay_marriage:party_closeRefuse to say/No answer	-0.062
56	issuerights_indep_integration:party_closeRefuse to say/No answer	-0.118
57	issueclimate_concern:party_closeSPD	0.012
58	issueeast_germans:party_closeSPD	-0.02
59	issueecon_inequality:party_closeSPD	0.016
60	issuegay_marriage:party_closeSPD	0.031
61	issuerights_indep_integration:party_closeSPD	-0.052

	std.error	p.value
	<dbl>	<dbl>

1	0.014	0
2	0.015	0
3	0.015	0.229
4	0.015	0.016
5	0.015	0
6	0.015	0
7	0.019	0.037
8	0.032	0.483
9	0.018	0.035
10	0.032	0.061
11	0.022	0.042
12	0.025	0.918
13	0.038	0.029
14	0.054	0.011
15	0.02	0.731
16	0.004	0.099
17	0.021	0.616
18	0.021	0.609
19	0.021	0.498
20	0.021	0.009
21	0.021	0.202
22	0.036	0.237
23	0.036	0.523
24	0.036	0.445
25	0.036	0.579
26	0.036	0.857
27	0.02	0.255
28	0.02	0.147
29	0.02	0.633
30	0.02	0.733
31	0.02	0.604
32	0.036	0.326
33	0.036	0.02
34	0.036	0.257
35	0.036	0.187
36	0.036	0.032
37	0.022	0
38	0.022	0.425
39	0.022	0.249
40	0.022	0
41	0.022	0.096
42	0.026	0
43	0.026	0.164
44	0.026	0
45	0.026	0
46	0.026	0
47	0.044	0.161
48	0.044	0.922
49	0.044	0.28
50	0.044	0.353
51	0.044	0.569
52	0.064	0.08
53	0.064	0.235

```
54      0.064  0.589
55      0.064  0.335
56      0.064  0.065
57      0.022  0.582
58      0.022  0.345
59      0.022  0.452
60      0.022  0.157
61      0.022  0.016
```

```
=== H6 (issue weights: corrP): party_close × issue interaction ===
      chi^2(54) = 392.980,  p = 0.0000  ***
```

Interpretation 2: Within each question, do parties differ?

```
run_H_per_issue("exp",      "party_close", "H6", use_lmer = FALSE, categorical = TRUE)
run_H_per_issue("corrP",   "party_close", "H6", use_lmer = TRUE,  categorical = TRUE)
```

```
-- H6 per issue (issue weight: exp) --
climate_concern  R^2=0.007  F(9,2118)=1.67  p=0.0902 .
gay_marriage     R^2=0.006  F(9,2118)=1.21  p=0.2839 ns
rights_indep_integration  R^2=0.008  F(9,2118)=1.75  p=0.0725 .
econ_inequality  R^2=0.006  F(9,2118)=1.35  p=0.2040 ns
regulate_internet R^2=0.007  F(9,2118)=1.57  p=0.1183 ns
east_germans     R^2=0.009  F(9,2118)=1.86  p=0.0534 .
-- H6 per issue (issue weight: corrP) --
climate_concern  R^2=0.062/0.557  LRT chi^2(9)=100.48  p=0.0000 ***
gay_marriage     R^2=0.072/0.535  LRT chi^2(9)=118.38  p=0.0000 ***
rights_indep_integration  R^2=0.049/0.540  LRT chi^2(9)=81.67  p=0.0000 ***
econ_inequality  R^2=0.035/0.475  LRT chi^2(9)=58.79  p=0.0000 ***
regulate_internet R^2=0.026/0.409  LRT chi^2(9)=45.72  p=0.0000 ***
east_germans     R^2=0.016/0.429  LRT chi^2(9)=27.01  p=0.0014 **
```

3.2.2 H7

3.2.2.1 H7. Left-Right placement predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot lr_n + u_q + \epsilon_{n,q} \quad (12)$$

with continuous variable lr_n between 'left' (-5) and 'right' (5). We estimate this model jointly across all six issues. β_0 is average. Because α is normalised: $\beta_0 = 1/6$ lr_n is explanatory variable u_q is average deviation of the weight for issue q ϵ_{nq} is individual specific error for this question and this individual.

The prediction is $\beta_q \neq 0$. Left and right participants differ in how strongly they weigh different political issues when mapping individuals. We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```
# H7 - lr
run_H_joint      ("linear", "lr", "H7", use_lmer = FALSE)
run_H_joint      ("corrP",  "lr", "H7", use_lmer = TRUE)
```

```
R^2  Null = 0.009  |  Full = 0.010
      term estimate std.error p.value
1      (Intercept)    0.129    0.011  0.000
2      issueclimate_concern    0.093    0.020  0.000
3      issueeast_germans    0.005    0.017  0.786
4      issueecon_inequality    0.021    0.017  0.208
5      issuegay_marriage    0.046    0.018  0.009
6      issuerights_indep_integration    0.062    0.019  0.001
7      lr    0.014    0.021  0.495
8      wave2    0.000    0.000  0.779
9      issueclimate_concern:lr   -0.068    0.039  0.081
10     issueeast_germans:lr    0.044    0.032  0.167
11     issueecon_inequality:lr   0.012    0.032  0.704
12     issuegay_marriage:lr    -0.062    0.033  0.059
13     issuerights_indep_integration:lr   -0.012    0.036  0.733
=== H7 (issue weights: linear): lr × issue interaction ===
```

```

F(6, 12761) = 3.087,  p = 0.0051  **

Model      AIC      BIC logLik npar delta_AIC delta_BIC
Null -2358.8 -2291.7 1188.4   7     186.6     141.9
Full -2545.4 -2433.6 1287.7  13        0.0        0.0
# A tibble: 13 x 4
  term                                estimate std.error p.value
  <chr>                                <dbl>     <dbl>   <dbl>
1 (Intercept)                        0.201      0.015     0
2 issueclimate_concern                0.221      0.015     0
3 issueeast_germans                    0         0.015   0.989
4 issueecon_inequality                 0.124      0.015     0
5 issuegay_marriage                    0.218      0.015     0
6 issuerights_indep_integration        0.154      0.015     0
7 lr                                   -0.002     0.029   0.945
8 wave2                               0.008      0.004   0.023
9 issueclimate_concern:lr             -0.209     0.029     0
10 issueeast_germans:lr                0.035     0.029   0.23
11 issueecon_inequality:lr            -0.152     0.029     0
12 issuegay_marriage:lr               -0.269     0.029     0
13 issuerights_indep_integration:lr   -0.063     0.029   0.03
=== H7 (issue weights: corrP): lr × issue interaction ===
    chi^2(6) = 198.683,  p = 0.0000  ***

```

And again, per issue:

```

run_H_per_issue("exp",    "lr", "H7", use_lmer = FALSE)
run_H_per_issue("corrP",  "lr", "H7", use_lmer = TRUE)

```

```

-- H7 per issue (issue weight: exp) --
climate_concern  R^2=0.002  F(1,2126)=4.96  p=0.0260 *
gay_marriage     R^2=0.004  F(1,2126)=5.28  p=0.0217 *
rights_indep_integration  R^2=0.000  F(1,2126)=0.54  p=0.4634 ns
econ_inequality  R^2=0.001  F(1,2126)=2.69  p=0.1014 ns
regulate_internet R^2=0.001  F(1,2126)=1.19  p=0.2764 ns
east_germans     R^2=0.002  F(1,2126)=1.62  p=0.2028 ns
-- H7 per issue (issue weight: corrP) --
climate_concern  R^2=0.031/0.567  LRT chi^2(1)=50.01  p=0.0000 ***
gay_marriage     R^2=0.047/0.530  LRT chi^2(1)=77.36  p=0.0000 ***
rights_indep_integration  R^2=0.005/0.547  LRT chi^2(1)=7.66  p=0.0056 **
econ_inequality  R^2=0.021/0.479  LRT chi^2(1)=33.64  p=0.0000 ***
regulate_internet R^2=0.001/0.404  LRT chi^2(1)=0.52  p=0.4728 ns
east_germans     R^2=0.000/0.427  LRT chi^2(1)=0.00  p=0.9496 ns

```

3.2.3 H8

3.2.3.1 H8. Age predicts issue weights.

We model the issue weights for each issue q as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot age_n + u_q + \epsilon_{n,q} \quad (13)$$

The prediction is $\beta_q \neq 0$. Older and younger participants differ in how strongly they weigh different political issues when mapping individuals.

```

run_H_joint      ("linear", "age", "H8", use_lmer = FALSE)
run_H_joint      ("corrP",  "age", "H8", use_lmer = TRUE)

```

```

R^2  Null = 0.009  |  Full = 0.011
      term estimate std.error p.value
1      (Intercept)  0.107     0.015  0.000
2  issueclimate_concern  0.034     0.026  0.191
3  issueeast_germans    0.053     0.025  0.035
4  issueecon_inequality  0.083     0.024  0.001
5  issuegay_marriage    0.070     0.024  0.004

```

```

6      issuerights_indep_integration    0.118    0.026    0.000
7              age                    0.001    0.000    0.048
8              wave2                   0.000    0.000    0.825
9      issueclimate_concern:age        0.001    0.001    0.304
10     issueeast_germans:age          -0.001    0.000    0.262
11     issueecon_inequality:age       -0.001    0.000    0.016
12     issuegay_marriage:age          -0.001    0.000    0.020
13 issuerights_indep_integration:age  -0.001    0.001    0.010
=== H8 (issue weights: linear): age × issue interaction ===
      F(6, 12659) = 4.416,  p = 0.0002  ***

```

```

Model      AIC    BIC logLik npar delta_AIC delta_BIC
Null -2319.0 -2252 1168.5    7      27.7      0
Full -2346.7 -2235 1188.3   13       0.0     17
# A tibble: 13 x 4
  term                                estimate std.error p.value
  <chr>                                <dbl>     <dbl>   <dbl>
1 (Intercept)                        0.159     0.022     0
2 issueclimate_concern                0.154     0.021     0
3 issueeast_germans                   0.005     0.021   0.807
4 issueecon_inequality                0.112     0.021     0
5 issuegay_marriage                   0.159     0.021     0
6 issuerights_indep_integration        0.209     0.021     0
7 age                                0.001      0      0.051
8 wave2                              0.008     0.004   0.041
9 issueclimate_concern:age            -0.001      0      0.091
10 issueeast_germans:age               0         0      0.553
11 issueecon_inequality:age           -0.001      0      0.002
12 issuegay_marriage:age              -0.001      0         0
13 issuerights_indep_integration:age  -0.002      0         0
=== H8 (issue weights: corrP): age × issue interaction ===
      chi^2(6) = 39.647,  p = 0.0000  ***

```

And again, per issue:

```

run_H_per_issue("exp",    "age", "H8", use_lmer = FALSE)
run_H_per_issue("corrP",  "age", "H8", use_lmer = TRUE)

```

```

-- H8 per issue (issue weight: exp) --
climate_concern  R^2=0.004 F(1,2109)=8.33 p=0.0039 **
gay_marriage     R^2=0.002 F(1,2109)=2.17 p=0.1405 ns
rights_indep_integration R^2=0.002 F(1,2109)=2.62 p=0.1059 ns
econ_inequality  R^2=0.001 F(1,2109)=2.52 p=0.1128 ns
regulate_internet R^2=0.001 F(1,2109)=1.38 p=0.2396 ns
east_germans     R^2=0.001 F(1,2109)=0.49 p=0.4847 ns
-- H8 per issue (issue weight: corrP) --
climate_concern  R^2=0.001/0.570 LRT chi^2(1)=0.01 p=0.9344 ns
gay_marriage     R^2=0.001/0.537 LRT chi^2(1)=1.76 p=0.1844 ns
rights_indep_integration R^2=0.003/0.549 LRT chi^2(1)=3.70 p=0.0545 .
econ_inequality  R^2=0.001/0.484 LRT chi^2(1)=0.74 p=0.3901 ns
regulate_internet R^2=0.003/0.406 LRT chi^2(1)=4.11 p=0.0426 *
east_germans     R^2=0.005/0.427 LRT chi^2(1)=7.42 p=0.0064 **

```

3.2.4 Exploration

3.2.4.1 Explorative Analyses.

We also model **H6** to **H8** with different combinations of party preference, left-right placement, and age of the participants to model participants issue weights $\alpha_{n,q}$. In addition, we model how political interest, pi_n , pf_n , interacts with the explanatory variables above.

```
print(" exploration later ")
```

```
[1] " exploration later "
```


3.3 Aim 3

Aim 3: Assess the role of social circle belief variation

For Aim 3, we test whether including social circle belief variation improves predictions for participants' issue weights (inferred from the map distances), their reported issue importance, likeability of voters, and polarisation perceptions. In addition, we explore how social circle belief improves predictions of the map distances between individuals.

3.3.1 H9

3.3.1.1 H9. Social circle variance predicts issue weights.

We model issue weights of participant n as

$$\alpha_{n,q} = 1/6 + \beta_q \cdot \text{var}_{n,q} + u_q + \epsilon_{n,q} . \quad (14)$$

The prediction is $\beta_q < 0$. Issues are assigned higher weight when associated with lower social circle variance.

CHANGE TO PREREG: I am using standard deviation instead of variance now!!!

$$\alpha_{n,q} = 1/6 + \beta_q \cdot \text{var}_{n,q} + u_q + \epsilon_{n,q} . \quad (15)$$

```
run_h9 <- function(k = "exp", variance_type = "std") {
  var_col <- paste0(variance_type, "_socialCircle_ops")

  df_h9 <- df_p |>
    filter(wave %in% c(1, 2)) |>
    pivot_longer(
      cols      = matches(paste0("^(", var_col, "|", k, "_alpha)_")),
      names_to   = c(".value", "issue"),
      names_pattern = paste0("^(", var_col, "|", k, "_alpha)_(.*)")
    ) |>
    rename(var_sc = all_of(var_col), alpha = all_of(paste0(k, "_alpha"))) |>
    filter(!is.na(alpha), !is.na(var_sc))

  m <- lmer(alpha ~ var_sc + wave + (1 | issue), data = df_h9)
  cf <- lmer_coef(m, "var_sc")
  print_one_sided(
    sprintf("H9: Social circle %s predicts issue weights (%s)", variance_type, k),
    cf$beta, cf$se, cf$t, direction = "negative"
  )

  # Per-issue
  cat("Per-issue breakdown:\n")
  purrr::walk(questions_sc, function(q) {
    df_q <- df_p |>
      filter(wave %in% c(1, 2)) |>
      select(id, wave,
        var_sc = all_of(paste0(var_col, "_", q)),
        alpha = all_of(paste0(k, "_alpha_", q))) |>
      filter(!is.na(alpha), !is.na(var_sc))
    m_q <- lmer(alpha ~ var_sc + wave + (1 | id), data = df_q)
    cf_q <- lmer_coef(m_q, "var_sc")
    cat(sprintf("  %-35s beta=%7.4f t=%6.3f p=%.4f %s\n",
      q, cf_q$beta, cf_q$t, cf_q$p2, sig_stars(cf_q$p2)))
  })
  cat("\n")
}
```

```
run_h9("exp", "std")
run_h9("corrP", "std")
```

```
=== H9: Social circle std predicts issue weights (exp) ===
Estimate = -0.0788
SE        =  0.0098
t         = -8.0041
df        = infinity
```

```
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

Per-issue breakdown:

climate_concern	beta=-0.0668	t=-2.659	p=0.0079	**
gay_marriage	beta=-0.0807	t=-3.665	p=0.0003	***
rights_indep_integration	beta=-0.0161	t=-0.648	p=0.5169	ns
econ_inequality	beta=-0.0506	t=-1.866	p=0.0622	.
regulate_internet	beta=-0.1390	t=-6.158	p=0.0000	***
east_germans	beta=-0.1574	t=-5.752	p=0.0000	***

=== H9: Social circle std predicts issue weights (corrP) ===

```
Estimate = -0.1352
SE        = 0.0116
t         = -11.7050
df        = infty
p (one-sided, beta < 0) = 0.0000
Supported : YES
```

Per-issue breakdown:

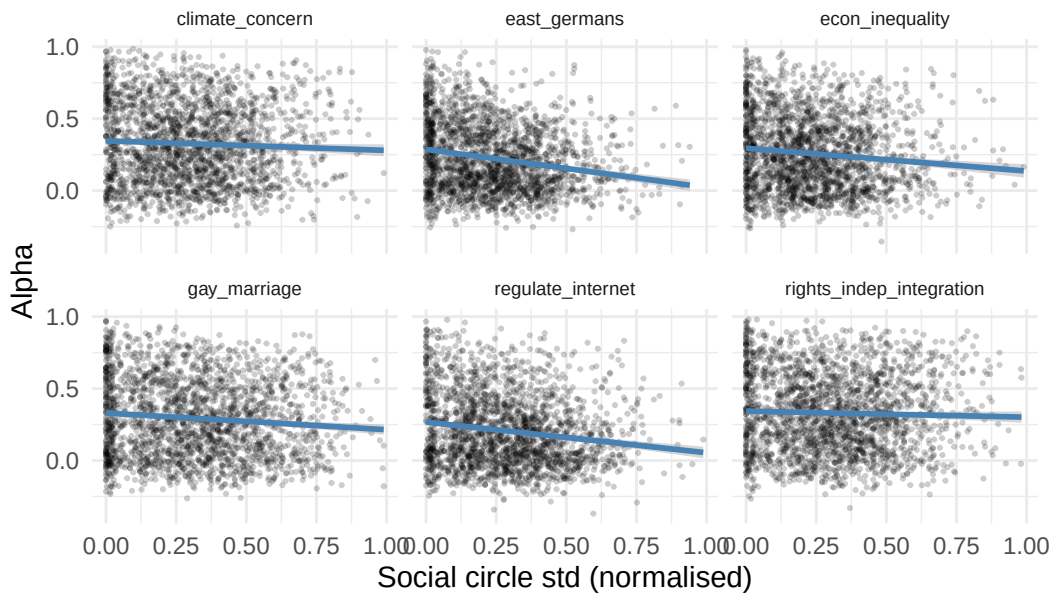
climate_concern	beta=-0.0997	t=-3.373	p=0.0008	***
gay_marriage	beta=-0.1451	t=-5.472	p=0.0000	***
rights_indep_integration	beta=-0.0746	t=-2.633	p=0.0085	**
econ_inequality	beta=-0.1673	t=-5.342	p=0.0000	***
regulate_internet	beta=-0.2196	t=-7.577	p=0.0000	***
east_germans	beta=-0.2537	t=-8.779	p=0.0000	***

```
k <- "corrP"; variance_type <- "std"
df_h9_plot <- df_p |>
  filter(wave %in% c(1, 2)) |>
  pivot_longer(
    cols      = matches(paste0("^ (std_socialCircle_ops|", k, "_alpha)_")),
    names_to   = c(".value", "issue"),
    names_pattern = paste0("^ (std_socialCircle_ops|", k, "_alpha)_(.*)" )
  ) |>
  rename(var_sc = all_of(paste0(variance_type, "_socialCircle_ops")), alpha = all_of(paste0(k, "_alpha"))) |>
  filter(!is.na(alpha), !is.na(var_sc))

ggplot(df_h9_plot, aes(x = var_sc, y = alpha)) +
  geom_point(alpha = 0.2, size = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "steelblue") +
  facet_wrap(~ issue, ncol = 3) +
  labs(title = sprintf("H9: Social circle %s vs %s alpha", variance_type, k),
       x = sprintf("Social circle %s (normalised)", variance_type), y = "Alpha") +
  theme_minimal() +
  theme(strip.text = element_text(size = 7))

`geom_smooth()` using formula = 'y ~ x'
```

H9: Social circle std vs corrP alpha



```
ggsave(sprintf("figs/h9_plot_%s_%s.png", k, variance_type), width = 16/2.54, height = 9/2.54)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

3.3.2 H10

3.3.2.1 H10. Social circle variance predicts reported issue importance.

We model reported issue importance of participant n as

$$w_{n,q} = \beta_0 + \beta_q \cdot \text{var}_{n,q} + u_q + v_n + \epsilon_{n,q} . \quad (16)$$

where v_n is the average importance of all issues assigned by participant n .

The prediction is $\beta_q < 0$. Issues are rated as more important when social circle variance is smaller.

```
run_h10 <- function(variance_type = "std") {
  var_col <- paste0(variance_type, "_socialCircle_ops")

  df_h10 <- df_p |>
    filter(wave %in% c(1, 2)) |>
    pivot_longer(
      cols = c(all_of(paste0("w_", questions_sc)),
               all_of(paste0(var_col, "_", questions_sc))),
      names_to = c(".value", "issue"),
      names_pattern = paste0("^(", var_col, "|w)_(.*)")
    ) |>
    rename(var_sc = all_of(var_col))

  m_h10 <- lmer(w ~ var_sc + wave + (1 | issue) + (1 | id), data = df_h10)
  cf_h10 <- lmer_coef(m_h10, "var_sc")

  print_one_sided(
    sprintf("H10: Social circle %s predicts reported issue importance", variance_type),
    cf_h10$beta, cf_h10$se, cf_h10$t, direction = "negative"
  )

  # Per-issue
  cat("Per-issue breakdown:\n")
  purrr::walk(questions_sc, function(q) {
    df_q <- df_h10 |> filter(issue == q) |> select(id, wave, issue, w, var_sc)
    m_q <- lmer(w ~ var_sc + wave + (1 | id), data = df_q)
    cf_q <- lmer_coef(m_q, "var_sc")
    cat(sprintf(" %-35s beta=%7.4f t=%6.3f p=%4f %s\n",
                q, cf_q$beta, cf_q$t, cf_q$p2, sig_stars(cf_q$p2)))
  })
}
```

```

})

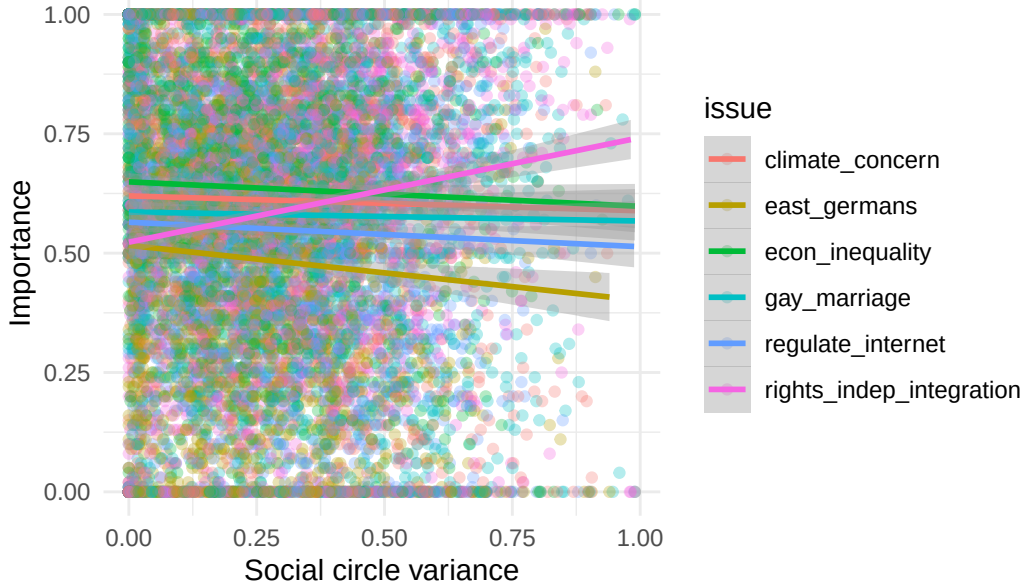
ggplot(df_h10, aes(x = var_sc, y = w, color = issue)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "Social circle variance", y = "Importance",
       title = sprintf("H10: Social circle %s vs relative importance", variance_type)) +
  theme_minimal()
}

run_h10(variance_type="std")

```

`geom\_smooth()` using formula = 'y ~ x'

H10: Social circle std vs relative importance



=== H10: Social circle std predicts reported issue importance ===

```

Estimate = 0.0053
SE       = 0.0141
t        = 0.3748
df       = infty
p (one-sided, beta < 0) = 0.6461
Supported : NO

```

Per-issue breakdown:

climate_concern	beta=-0.0079	t=-0.260	p=0.7952	ns
gay_marriage	beta=-0.0176	t=-0.589	p=0.5559	ns
rights_indep_integration	beta= 0.1817	t= 5.917	p=0.0000	***
econ_inequality	beta=-0.0584	t=-1.919	p=0.0551	.
regulate_internet	beta=-0.0528	t=-1.704	p=0.0885	.
east_germans	beta=-0.1060	t=-3.146	p=0.0017	**

3.3.3 H11

3.3.3.1 H11. Overall social circle variance helps predict likeability rating of typical voters.

In H2 (equation 2), we modelled the participant n 's likeability rating of the typical voter of political party r as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + u_n + \epsilon_{n,r}. \quad (17)$$

Here, we extend this model as

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} + v_n + \epsilon_{n,r} \quad (18)$$

where the new β_2 -term represents the influence of overall social circle belief variation on likeability.

The prediction is $\beta_2 > 0$ and that equation 19 improves the predictions of the baseline model from **H2** (equation 2/17) using R2, AIC, and BIC as model quality indicators. Voters are more liked in general if the participant's overall social circle variance is higher.

TODO: This hypothesis is kind of wrong. We have 1 avgsocialcirclestd per person and 7 pixeldist and sympathy for 7 parties. But the avgsocialcirclestd is already included in the random intercept.

$$y_{n,r} = \beta_0 + \beta_1 \cdot d_{n,\text{self}-r} + \beta_2 \cdot \sum_q \text{var}_{n,q} \text{REMOVED}(+v_n) + \epsilon_{n,r} \quad (19)$$

```
variance_type <- "std"

avg_var_sc <- df_p |>
  filter(wave == 2) |>
  select(id, all_of(paste0(variance_type, "_socialCircle_ops_", questions_sc))) |>
  drop_na() |>
  mutate(avg_var_sc = rowMeans(across(-id))) |>
  select(id, avg_var_sc)

df_h11 <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, sympathy, party, pixel_dist) |>
  left_join(avg_var_sc, by = "id") |>
  mutate(
    # pixel_dist_z = as.numeric(scale(pixel_dist)),
    avg_var_sc_z = as.numeric(scale(avg_var_sc))

m_base_h11 <- lmer(sympathy ~ pixel_dist + (1 | id),
  data = df_h11, REML = FALSE)
m_ext_h11 <- lmer(sympathy ~ pixel_dist + avg_var_sc_z + (1 | id),
  data = df_h11, REML = FALSE)

cf_h11 <- lmer_coef(m_ext_h11, "avg_var_sc_z")

# One-sided test: H1 = avg_var_sc_z has a positive effect
p_one_h11 <- if (cf_h11$beta > 0) {
  cf_h11$p2 / 2
} else {
  1 - cf_h11$p2 / 2
}

r2_base <- r2(m_base_h11)
r2_ext <- r2(m_ext_h11)

cat("=== H11: Social circle variance predicts likeability (beyond map distance) ===\n")
cat(sprintf(" beta (pixel_dist) = %.4f\n",
  fixef(m_ext_h11)["pixel_dist"]))
cat(sprintf(" beta (avg_var_sc_z) = %.4f\n",
  cf_h11$beta))
cat(sprintf(" SE = %.4f\n",
  cf_h11$se))
cat(sprintf(" t = %.4f\n",
  cf_h11$t))
cat(sprintf(" df (Satterthwaite) = %.2f\n",
  cf_h11$df))
cat(sprintf(" p (two-sided) = %.6f\n",
  cf_h11$p2))
cat(sprintf(" p (one-sided, beta > 0) = %.6f\n",
  p_one_h11))
cat(sprintf(" Supported: %s\n",
  ifelse(p_one_h11 < 0.05 && cf_h11$beta > 0,
    "YES", "NO")))
cat(sprintf(" R² marginal base/ext = %.4f / %.4f (Delta=+.4f)\n",
  r2_base$R2_marginal,
```

```

      r2_ext$R2_marginal,
      r2_ext$R2_marginal - r2_base$R2_marginal))
cat(sprintf(" R² conditional base/ext = %.4f / %.4f (Delta=%+.4f)\n",
      r2_base$R2_conditional,
      r2_ext$R2_conditional,
      r2_ext$R2_conditional - r2_base$R2_conditional))
cat(sprintf(" AIC base/ext = %.2f / %.2f (Delta=%+.2f)\n",
      AIC(m_base_h11),
      AIC(m_ext_h11),
      AIC(m_ext_h11) - AIC(m_base_h11)))
cat(sprintf(" BIC base/ext = %.2f / %.2f (Delta=%+.2f)\n\n",
      BIC(m_base_h11),
      BIC(m_ext_h11),
      BIC(m_ext_h11) - BIC(m_base_h11)))

```

=== H11: Social circle variance predicts likeability (beyond map distance) ===

```

beta (pixel_dist) = -1.1115
beta (avg_var_sc_z) = -0.0135
SE = 0.0052
t = -2.5813
df (Satterthwaite) = 854.38
p (two-sided) = 0.010010
p (one-sided, beta > 0) = 0.994995
Supported: NO
R² marginal base/ext = 0.4208 / 0.4211 (Delta=+0.0002)
R² conditional base/ext = 0.5802 / 0.5775 (Delta=-0.0027)
AIC base/ext = 313.82 / 303.83 (Delta=-9.99)
BIC base/ext = 341.08 / 337.89 (Delta=-3.19)

```

3.3.4 H12a

3.3.4.1 H12a. Social circle variance predicts perceived political polarisation on a given issue.

We model perceived polarisation on issue q as

$$P_{n,q} = \beta_0 + \beta_q \cdot var_{n,q} + u_n + \epsilon_{n,q}. \quad (20)$$

The prediction is $\beta_q < 0$. Participants with less variation in their social circles with respect to issue q perceive higher polarisation on that issue. We test this via a single F-test on the joint significance of $\{\beta_q\}$, with significance threshold $\alpha = 0.05$.

```

variance_type <- "std"
var_col <- paste0(variance_type, "_socialCircle_ops")

df_h12a <- df_p |>
  filter(wave %in% c(1, 2)) |>
  select(-P_tot) |>
  pivot_longer(
    cols = c(all_of(paste0("P_", questions_sc)),
              all_of(paste0(var_col, "_", questions_sc))),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, "|P)_(.*)" ) |>
  rename(var_sc = all_of(var_col))

m_h12a <- lmer(P ~ var_sc + wave + (1 | issue) + (1 | id), data = df_h12a)
cf_h12a <- lmer_coef(m_h12a, "var_sc")

print_one_sided(
  sprintf("H12a: Social circle %s predicts perceived polarisation (per issue)", variance_type),
  cf_h12a$beta, cf_h12a$se, cf_h12a$t, direction = "negative"
)

ggplot(df_h12a, aes(x = var_sc, y = P, color = issue)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +

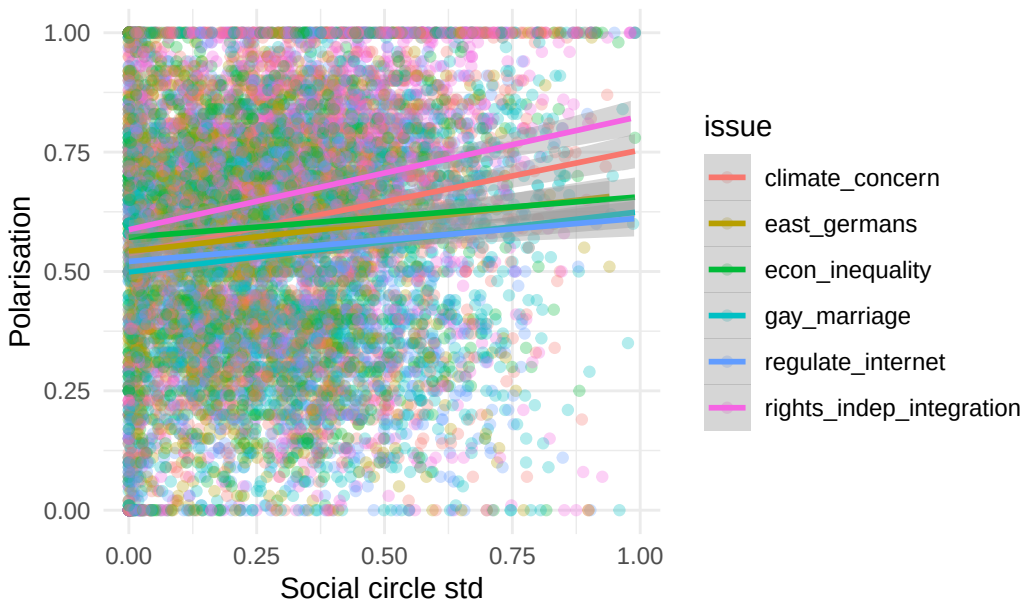
```



```
labs(x = sprintf("Social circle %s", variance_type), y = "Polarisation",
     title = sprintf("H12a: Social circle %s vs issue polarisation", variance_type)) +
theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

H12a: Social circle std vs issue polarisation



```
=== H12a: Social circle std predicts perceived polarisation (per issue) ===
```

```
Estimate = 0.1685
SE       = 0.0117
t        = 14.4028
df       = inf
p (one-sided, beta < 0) = 1.0000
Supported : NO
```

3.3.5 H12b

3.3.5.1 H12b. Social circle variance predicts perceived political polarisation.

We model overall perceived polarisation as

$$P_{n,tot} = \beta_0 + \beta_1 \cdot \sum_q var_{n,q} + \epsilon_n. \quad (21)$$

The prediction is $\beta_1 < 0$. Participants with less variation in their social circles averaged over all issues perceive more political polarisation. We test this via a t-test with significance threshold $\alpha = 0.05$.

```
df_h12b <- df_p |>
  select(id, wave, all_of(paste0(variance_type, "_socialCircle_ops_"), questions_sc), P_tot) |>
  drop_na() |>
  mutate(avg_var_sc = rowSums(across(all_of(paste0(variance_type, "_socialCircle_ops_"), questions_sc))))

m_h12b <- lmer(P_tot ~ avg_var_sc + (1 | id), data = df_h12b, REML = TRUE)
n_p_h12b <- n_distinct(df_h12b$id)
cf_h12b <- lmer_coef(m_h12b, "avg_var_sc")
t_h12b <- cf_h12b$beta / cf_h12b$se

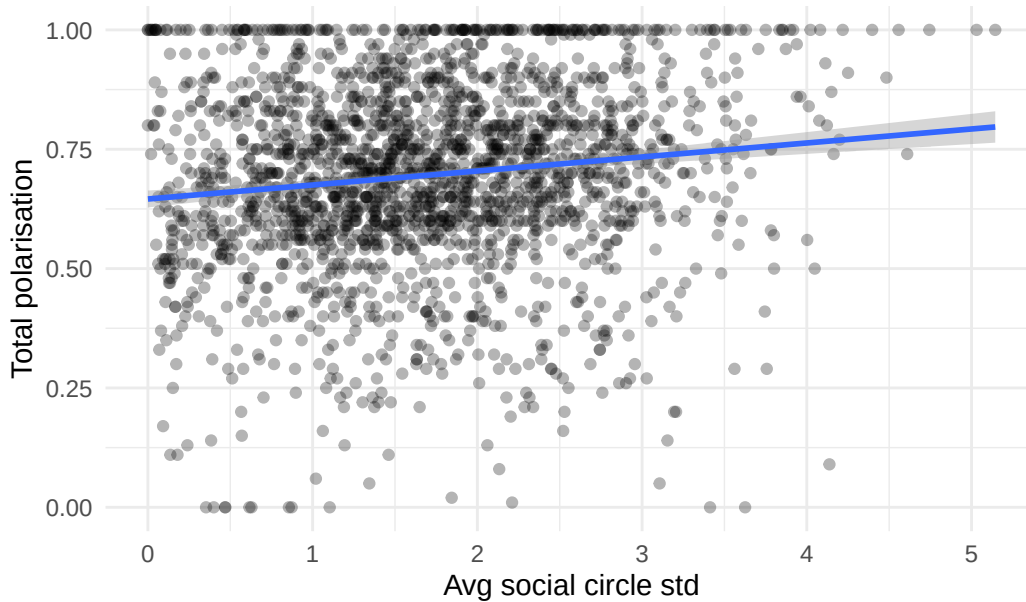
print_one_sided(
  "H12b: Overall social circle variance predicts total perceived polarisation",
  cf_h12b$beta, cf_h12b$se, t_h12b, df = n_p_h12b - 2,
  direction = "negative"
)

ggplot(df_h12b, aes(x = avg_var_sc, y = P_tot)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = sprintf("Avg social circle %s", variance_type), y = "Total polarisation",
```

```
title = sprintf("H12b: Social circle %s vs total polarisation", variance_type)) +
theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

H12b: Social circle std vs total polarisation



```
=== H12b: Overall social circle variance predicts total perceived polarisation ===
```

```
Estimate = 0.0234
SE        = 0.0050
t         = 4.6493
df        = 1225.0
p (one-sided, beta < 0) = 1.0000
Supported : NO
```

3.3.6 H13

3.3.6.1 H13. Reported issue importance predicts perceived distance.

In equation ??, we fix the issue weights to $\alpha_{n,q} = w_{n,q}$. Then, we model the perceived distance between two individuals i and j (using the exponential kernel; see equations ?? and ??) as

$$d_{n,ij} = 1 - \exp \left(-\hat{b}_n \cdot \left(\sum_q w_{n,q} \cdot |x_{n,i,q} - x_{n,j,q}| \right)^{\hat{c}_n} \right) \quad (22)$$

with fit parameters $\hat{b}_n$, and $\hat{c}_n$.

The prediction is that this model (with $\hat{\alpha}_{n,q} = w_{n,q}$) outperforms (i) an unweighted model with $\hat{\alpha}_{n,q} = 1/6$ and (ii) a population weight model with $\hat{\alpha}_{n,q} = 1/n \cdot \sum_n w_{n,q}$. We use R2, AIC, and BIC as model quality indicators.

```
linear_kernel <- function(x, a, b) a + b * x
exp_kernel    <- function(x, a, b) 1 - exp(-a * (x ^ b))
r2_vec        <- function(y, pm) {
  ss_tot <- sum((y - mean(y, na.rm = TRUE))^2, na.rm = TRUE)
  1 - colSums((pm - y)^2, na.rm = TRUE) / ss_tot
}
compute_aic_bic <- function(mse, k, n) {
  rss <- mse * n
  tibble(aic = n * log(rss / n) + 2 * k, bic = n * log(rss / n) + k * log(n))
}

alphas_pop_lin <- df_p |> select(all_of(col_alpha_lin)) |> colMeans()
alphas_pop_exp <- df_p |> select(all_of(col_alpha_exp)) |> colMeans()

df_h13_base <- df_diff |>
  filter(wave %in% c(1, 2)) |>
```



```

left_join(df_p |> select(id, wave, all_of(col_weights),
                        all_of(col_alpha_lin), all_of(col_alpha_exp)),
          by = c("id", "wave"), suffix = c("", ".drop")) |>
select(-ends_with(".drop"))

.d      <- as.matrix(df_h13_base[, col_deltas])
.a_uniform <- matrix(1 / length(questions_sc), nrow = nrow(df_h13_base), ncol = length(questions_sc))
.a_pop_exp <- matrix(alphas_pop_exp, nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_pop_lin <- matrix(alphas_pop_lin, nrow = nrow(df_h13_base), ncol = length(questions_sc), byrow = TRUE)
.a_fit_exp <- as.matrix(df_h13_base[, col_alpha_exp])
.a_fit_lin <- as.matrix(df_h13_base[, col_alpha_lin])
.a_weights <- {
  w <- as.matrix(df_h13_base[, col_weights])
  w_sum <- rowSums(w, na.rm = TRUE)
  ifelse(w_sum == 0, 0, w / w_sum)
}

pred_cols <- c("predicted_d_weights", "predicted_d_population_exp",
              "predicted_d_population_lin", "predicted_d_uniform_exp",
              "predicted_d_uniform_lin", "predicted_d_fit_linear",
              "predicted_d_fit_exp")

df_h13 <- df_h13_base |>
mutate(
  predicted_d_uniform_exp = exp_kernel( rowSums(.d * .a_uniform), exp_param1, exp_param2),
  predicted_d_uniform_lin = linear_kernel(rowSums(.d * .a_uniform), linear_param1, linear_param2),
  predicted_d_population_exp = exp_kernel( rowSums(.d * .a_pop_exp), exp_param1, exp_param2),
  predicted_d_population_lin = linear_kernel(rowSums(.d * .a_pop_lin), linear_param1, linear_param2),
  predicted_d_fit_exp = exp_kernel( rowSums(.d * .a_fit_exp), exp_param1, exp_param2),
  predicted_d_fit_linear = linear_kernel(rowSums(.d * .a_fit_lin), linear_param1, linear_param2),
  predicted_d_weights = exp_kernel( rowSums(.d * .a_weights), exp_param1, exp_param2)
) |>
mutate(across(all_of(pred_cols),
              ~ pixel_dist - .x,
              .names = "err_{str_remove(.col, 'predicted_d_')}"))

y      <- df_h13$pixel_dist
n_obs  <- sum(!is.na(y))
n_pw   <- df_h13 |> distinct(id, wave) |> nrow()
n_q    <- length(questions_sc)
pred_mat <- df_h13 |> select(all_of(pred_cols)) |> as.matrix()
colnames(pred_mat) <- str_remove(pred_cols, "predicted_d_")

k_lookup <- tribble(
  ~model,      ~k,
  "weights",    2,
  "population_exp", 2 + n_q,
  "population_lin", 2 + n_q,
  "uniform_exp",  2,
  "uniform_lin",  2,
  "fit_exp",      (2 + n_q) * n_pw,
  "fit_linear",   (2 + n_q) * n_pw
)
baseline_sq <- (pred_mat[, "uniform_exp"] - y)^2

comparison_tbl <- tibble(
  model = colnames(pred_mat),
  mse   = colMeans((pred_mat - y)^2, na.rm = TRUE),
  rmse  = sqrt(mse),
  r2    = r2_vec(y, pred_mat)
) |>
mutate(
  rel_improve = (mse[model == "uniform_exp"] - mse) / mse[model == "uniform_exp"],

```

```

p_wilcox      = map_dbl(model, function(m) {
  if (m == "uniform_exp") return(NA_real_)
  wilcox.test((pred_mat[, m] - y)^2, baseline_sq, paired = TRUE)$p.value
}),
mean_diff_vs_uniform = map_dbl(model, function(m) {
  if (m == "uniform_exp") return(NA_real_)
  mean((pred_mat[, m] - y)^2 - baseline_sq, na.rm = TRUE)
})
) |>
left_join(k_lookup, by = "model") |>
mutate(
  aic_bic = map2(mse, k, ~ compute_aic_bic(.x, .y, n_obs)),
  aic      = map_dbl(aic_bic, "aic"),
  bic      = map_dbl(aic_bic, "bic")
) |>
select(-aic_bic, -k) |>
arrange(mse)

print(comparison_tbl)

```

```

# A tibble: 7 x 9
  model      mse rmse      r2 rel_improve p_wilcox
<chr>      <dbl> <dbl>   <dbl>      <dbl>      <dbl>
1 fit_linear 0.0162 0.127 0.509      0.237      0
2 fit_exp    0.0164 0.128 0.505      0.230      0
3 population_exp 0.0212 0.146 0.358      0.00238 2.54e-71
4 uniform_exp 0.0213 0.146 0.357      0         NA
5 weights    0.0330 0.182 0.00175 -0.552      0
6 uniform_lin 0.0431 0.208 -0.301     -1.02     4.13e-14
7 population_lin 0.0466 0.216 -0.407     -1.19     1.29e- 2

  mean_diff_vs_uniform aic      bic
      <dbl>      <dbl>   <dbl>
1      -0.00505 -1102767. -923289.
2      -0.00490 -1100181. -920702.
3      -0.0000506 -1062758. -1062674.
4         NA      -1062113. -1062092.
5       0.0117    -940918. -940897.
6       0.0218    -867738. -867717.
7       0.0253    -846120. -846036.

```

3.3.6.2 Analysis

```

err_cols <- str_replace(pred_cols, "predicted_d_", "err_")

# Per-participant R^2
per_person_r2 <- df_h13 |>
  group_by(id) |>
  summarise(across(all_of(pred_cols), function(x) {
    v <- !is.na(x) & !is.na(pixel_dist)
    if (sum(v) < 3) return(NA_real_)
    ss_res <- sum((pixel_dist[v] - x[v])^2)
    ss_tot <- sum((pixel_dist[v] - mean(pixel_dist[v]))^2)
    if (ss_tot == 0) NA_real_ else 1 - ss_res / ss_tot
  }, .names = "r2_{.col}")) |>
  pivot_longer(starts_with("r2_"), names_to = "model", values_to = "r2") |>
  mutate(model = str_remove(model, "r2_predicted_d_"))

p1 <- ggplot(per_person_r2, aes(x = reorder(model, r2, median, na.rm = TRUE), y = r2)) +
  geom_boxplot(outlier.alpha = 0.8, outlier.size = 0.2) +
  stat_summary(fun = mean, geom = "point", shape = 15, size = 3, colour = "red") +
  geom_hline(yintercept = 0, colour = "red", linetype = "dashed") +
  coord_flip() + ylim(-2, 1) +
  labs(title = "H13: Per-participant R^2", x = NULL, y = expression(R^2)) +
  theme_minimal()

```

```
# Predicted vs observed
p2 <- df_h13 |>
  slice_sample(n = 10000) |>
  select(pixel_dist, all_of(pred_cols)) |>
  pivot_longer(-pixel_dist, names_to = "model", values_to = "predicted") |>
  mutate(model = str_remove(model, "predicted_d_")) |>
  ggplot(aes(pixel_dist, predicted)) +
  geom_point(alpha = 0.05, size = 0.3) +
  geom_abline(slope = 1, intercept = 0, colour = "red") +
  geom_smooth(method = "gam", formula = y ~ s(x, bs = "cs"), se = FALSE, colour = "steelblue") +
  facet_wrap(~ model, ncol = 4) + ylim(0, 1) + coord_equal() +
  labs(title = "H13: Predicted vs observed") + theme_minimal()

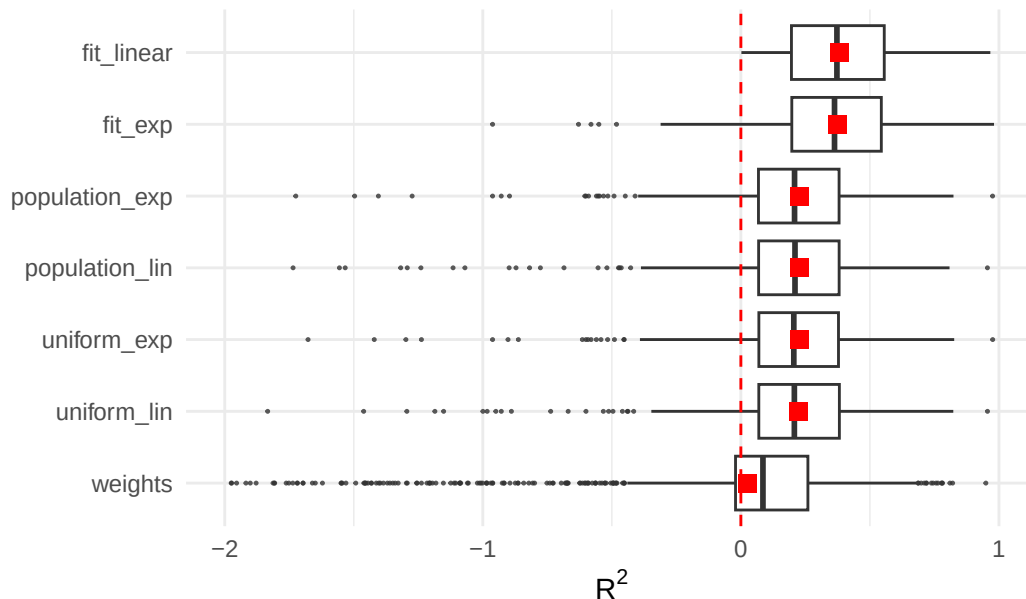
# Participant-level MSE
p3 <- df_h13 |>
  group_by(id) |>
  summarise(across(all_of(str_replace(pred_cols, "predicted_d_", "err_")),
    function(x) mean(x^2, na.rm = TRUE))) |>
  pivot_longer(-id, names_to = "model", values_to = "mse") |>
  mutate(model = str_remove(model, "^err_")) |>
  ggplot(aes(x = reorder(model, mse, median), y = mse)) +
  geom_boxplot(outlier.alpha = 0.15) +
  coord_flip() + ylim(0, 0.1) +
  labs(title = "H13: Participant-level MSE", x = NULL, y = "MSE") +
  theme_minimal()

p1; p2; p3
```

Warning: Removed 94 rows containing non-finite outside the scale range
(`stat\_boxplot()`).

Warning: Removed 94 rows containing non-finite outside the scale range
(`stat\_summary()`).

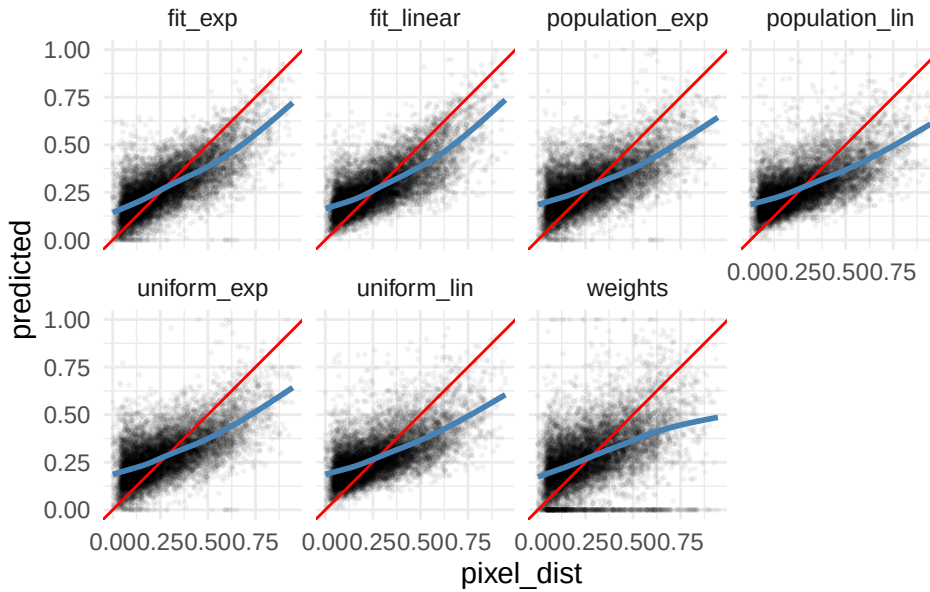
H13: Per-participant R^2



Warning: Removed 94 rows containing non-finite outside the scale range
(`stat\_smooth()`).

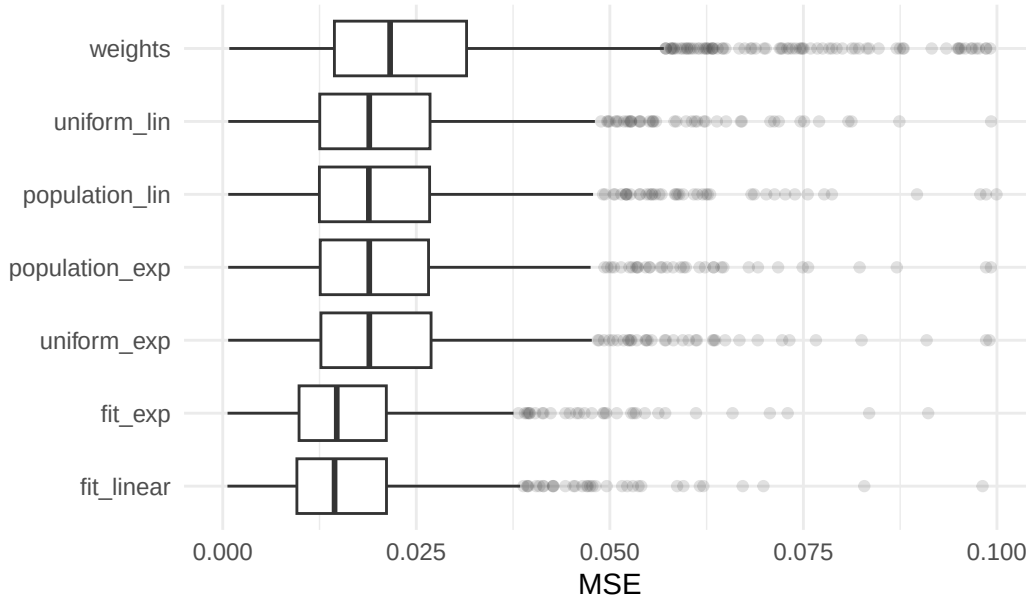
Warning: Removed 94 rows containing missing values or values outside the scale range
(`geom\_point()`).

H13: Predicted vs observed



Warning: Removed 115 rows containing non-finite outside the scale range (``stat_boxplot()``).

H13: Participant-level MSE



3.3.6.3 Explorative Analyses.

In addition to the preregistered hypotheses above, we explore several ways to define the function g_n (equation ??) which describes the (participant-level) aggregation of belief differences on the six items q between individuals i and j into an overall belief distance.

In particular, we model different functional forms of g_n based on theories about who consider as social context and how this social context shapes their perception of the belief space—which we refer to as a social *lens*.

Exploration of different lens models We define three different functional types of aggregation functions g_n with increasing level of complexity. First, a **simple lens** $\mathcal{S}_n$ is a 6-dimensional vector that scales (or weighs) each entry in the belief distance vector (equivalent to the weighted metric g_n in Box 1). Second, a **compression lens** $\mathcal{C}_n$ is a 6-dimensional vector, which projects the belief difference vector onto a single dimension, e.g. *liberal* vs. *conservative*, and scales the belief distance in the direction of this dimension. Third, a **full lens** $\mathcal{F}_n$ is a 6×6 matrix that transforms the belief distance vector into a subjective new space with the same number of dimensions but with rotated and scaled axes. Writing these lenses in terms of transformation matrices yields:

$$\begin{aligned}
1 \quad \text{simplified lens} \quad g_n(|x_{n,i,q} - x_{n,j,q}|) &= \left\| \begin{pmatrix} \mathcal{S}_n & 0 \\ 0 & \mathcal{S}_n \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p \\
2 \quad \text{compression lens} \quad g_n(|x_{n,i,q} - x_{n,j,q}|) &= \left\| \begin{pmatrix} -\mathcal{C}_n & - \\ 0 & \ddots & 0 \\ 0 & \vdots & 0 \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p \\
3 \quad \text{full lens} \quad g_n(|x_{n,i,q} - x_{n,j,q}|) &= \left\| \begin{pmatrix} \mathcal{F}_{n,11} & \dots & \mathcal{F}_{n,61} \\ \vdots & \dots & \vdots \\ \mathcal{F}_{n,61} & \dots & \mathcal{F}_{n,66} \end{pmatrix} \circ \begin{pmatrix} |x_{n,i,1} - x_{n,j,1}| \\ \vdots \\ |x_{n,i,6} - x_{n,j,6}| \end{pmatrix} \right\|_p
\end{aligned}$$

where $\|\cdot\|_p$ is the p-norm. Here, we choose Minkowski norm $p = 1$ or Euclidean norm $p = 2$.

The lenses $\mathcal{S}_n$, $\mathcal{C}_n$, and $\mathcal{F}_n$ of participant n depend on the belief distribution in the participant's social contexts, $x_{n,\text{context},q}$. We use two ways to define this social contexts:

- the beliefs of the seven to ten social circle contacts of the participant: $x_{n,\text{context},q} = \{x_{n,k,q} \mid k = 1 : K_n\}$.
- the beliefs of all the other participants in the participant's in-group, where in-group can mean (i) those who affiliate with the same party, $\{x_{m,\text{self},q} \mid m \text{ with party}_m = \text{party}_n\}$, (ii) those who place themselves in the same region on the left-right scale, (iii) those who have similar demographic attributes, or (iv) combinations thereof.

Then, we determine participant n 's lenses ...

- $\mathcal{S}_n$ as the inverse of the standard deviation of the $x_{n,\text{context},q}$. $\mathcal{S}_n = \left(\frac{1}{\sqrt{\text{Var}(x_{n,\text{context},1})}}, \dots, \frac{1}{\sqrt{\text{Var}(x_{n,\text{context},6})}} \right)$.
- $\mathcal{C}_n$ as the first (loaded) principal components of the social context beliefs. $\mathcal{C}_n = \frac{1}{\sqrt{\lambda_{n,1}}} \nu_{n,1}$.
- $\mathcal{F}_n$ as a principal component transformation, i.e as the inverse of the loaded principal component vectors of the social context beliefs. $\mathcal{F}_n = (\sqrt{\lambda_{n,1}} \nu_{n,1}, \dots, \sqrt{\lambda_{n,6}} \nu_{n,6})^{-1}$, where $\nu_{n,\tilde{q}}$ are principal components and $\lambda_{n,\tilde{q}}$ the corresponding eigenvalues.

We predict map distances $d_{n,ij}$ using equation ?? with different choices of lenses and different definitions of social contexts. The goal of this exploration is to investigate to what extent these (theoretical) models and assumptions about participant-level social lenses describe how people perceive political differences between individuals.

```
print("later")
```

```
[1] "later"
```

3.4 Aim 4

Aim 4: Test the effects of an intervention on social circle belief variation

In wave 2, participants are randomly put in (i) a control condition, where they receive the same instruction as in wave 1, namely to choose seven to ten social contacts that they would count to their social circle, or (ii) a treatment condition, where they receive an additional instruction to choose contacts such that they represent a diverse mix of beliefs, if possible. Aim 4 is to test the effects of this instruction.

3.4.1 H14

3.4.1.1 H14. Participants in the treatment condition report higher social circle variance.

For the participants in the treatment condition T , who receive the instruction to choose diverse contacts, we measure the increase in social circle variance on issue q as

$$\Delta \text{var}_T = \sum_q (\mathbb{E}[\text{var}_{n,q}(w2) \mid \text{condition}_n = T] - \mathbb{E}[\text{var}_{n,q}(w1) \mid \text{condition}_n = T]) \quad (23)$$

The prediction is $\Delta \text{var}_T > 0$ (one-sample t-test with significance threshold $\alpha = 0.05$) and $\Delta \text{var}_T > \Delta \text{var}_{\text{Control}}$ (using independent samples t-test with $\alpha = 0.05$). Social circle variance in the treatment condition T is higher than in wave 1 and higher than in the control group.

```

variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")

df_h14 <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(
    cols      = matches(paste0(var_col, "_")),
    names_to   = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, ")_(.*)")
  ) |>

```

```

pivot_wider(id_cols = c(id, issue), names_from = wave,
             names_prefix = "var_sc_wave", values_from = all_of(var_col)) |>
mutate(group = if_else(id %in% treatment_ids, "treatment", "control"),
       delta = var_sc_wave2 - var_sc_wave1)

cat("=== H14: Treatment increases social circle variance ===\n\n")

# Sanity check
df_h14 |>
  group_by(group) |>
  summarise(mean_w1 = mean(var_sc_wave1, na.rm = TRUE),
            sd_w1    = sd(var_sc_wave1,   na.rm = TRUE),
            mean_w2 = mean(var_sc_wave2, na.rm = TRUE),
            sd_w2    = sd(var_sc_wave2,   na.rm = TRUE),
            mean_d   = mean(delta,        na.rm = TRUE),
            n        = n()) |> print()

t_bal <- t.test(var_sc_wave1 ~ group, data = df_h14)
cat(sprintf(" Balance check (wave1 by group): t=%.3f  p=%.4f\n\n", t_bal$statistic, t_bal$p.value))

trt <- df_h14 |> filter(group == "treatment")
ctrl <- df_h14 |> filter(group == "control")

# H14-1: Deltavar_T > 0
t_h14_1 <- t.test(trt$delta, mu = 0, alternative = "greater")
cat(sprintf("H14-1 (Deltavar_treatment > 0):\n"))
cat(sprintf(" Mean delta = %.4f ± %.4f\n", mean(trt$delta, na.rm = TRUE), sd(trt$delta, na.rm = TRUE)))
cat(sprintf(" t = %.4f, p (one-sided) = %.4f\n", t_h14_1$statistic, t_h14_1$p.value))
cat(sprintf(" Supported : %s\n\n", ifelse(t_h14_1$statistic > 0 && t_h14_1$p.value < 0.05, "YES", "NO")))

# H14-2: Deltavar_T > Deltavar_Control
t_h14_2 <- t.test(delta ~ group, data = df_h14, alternative = "less", var.equal = FALSE)
cat(sprintf("H14-2 (Deltavar_treatment > Deltavar_control):\n"))
cat(sprintf(" Mean delta treatment = %.4f ± %.4f\n", mean(trt$delta, na.rm = TRUE), sd(trt$delta, na.rm = TRUE)))
cat(sprintf(" Mean delta control   = %.4f ± %.4f\n", mean(ctrl$delta, na.rm = TRUE), sd(ctrl$delta, na.rm = TRUE)))
cat(sprintf(" Difference (T - C)    = %.4f\n", mean(trt$delta, na.rm = TRUE) - mean(ctrl$delta, na.rm = TRUE)))
cat(sprintf(" t = %.4f, p (one-sided) = %.4f\n", t_h14_2$statistic, t_h14_2$p.value))
cat(sprintf(" Supported : %s\n\n", ifelse(t_h14_2$statistic < 0 && t_h14_2$p.value < 0.05, "YES", "NO")))

=== H14: Treatment increases social circle variance ===

# A tibble: 2 x 7
  group mean_w1 sd_w1 mean_w2 sd_w2 mean_d n
  <chr>   <dbl> <dbl>   <dbl> <dbl>   <dbl> <int>
1 control 0.268 0.198 0.275 0.197 0.00638 2370
2 treatment 0.285 0.199 0.311 0.205 0.0262 3042
  Balance check (wave1 by group): t=-3.044 p=0.0023

H14-1 (Deltavar_treatment > 0):
  Mean delta = 0.0262 ± 0.1884
  t = 7.6679, p (one-sided) = 0.0000
  Supported : YES

H14-2 (Deltavar_treatment > Deltavar_control):
  Mean delta treatment = 0.0262 ± 0.1884
  Mean delta control   = 0.0064 ± 0.1780
  Difference (T - C)    = 0.0198
  t = -3.9599, p (one-sided) = 0.0000
  Supported : YES

```

3.4.2 H15

3.4.2.1 H15. Higher relative social circle variance decreases issue weight.

We model the new issue weights, $\alpha_{n,q}(w2)$, inferred in wave 2, of the participants in the treatment condition as

$$\alpha_{n,q}(w2) = \alpha_{n,q}(w1) + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \frac{\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)}{\sum_q \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)} + \epsilon_{n,q} \quad (24)$$

The prediction is $\beta_1 < 0$. Higher relative social circle variance on one issue decreases the weight of that issue for the participant. %We test this via a t-test with significance threshold $\alpha = 0.05$.

UPDATE: There should be abs signs in the denominator! Otherwise small delta Var's are going to blow up things dramatically.

UPDATE: This does not need to depend on the Treatment condition group alone. I analysed both now

OPEN QUESTION: Is the denominator: $\sum_q |\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)|$ or $\sum_q |\text{var}_{n,q}(w1)|$? I chose the first now because this is what we preregistered.

```
variance_type <- "std"
k             <- "exp"
var_col       <- paste0(variance_type, "_socialCircle_ops")

# Variance change per participant per issue
df_h15_var <- df_p |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
               names_to = c(".value", "issue"),
               names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  rename(var_sc = all_of(var_col)) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "w", values_from = var_sc) |>
  rename(var_w1 = w1, var_w2 = w2) |>
  mutate(delta_var = var_w2 - var_w1) |>
  group_by(id) |>
  mutate(
    sum_delta_var = sum(abs(delta_var), na.rm = TRUE),
    # denominator is  $\Sigma |\text{Delta var}|$  (preregistered); see note in LaTeX about alternatives
    rel_delta_var = if_else(sum_delta_var > 0.05, delta_var / sum_delta_var, NA_real_)
  ) |>
  ungroup()

# Alpha weights across waves
df_h15_alpha <- df_p |>
  pivot_longer(cols = matches(paste0("^", k, "_alpha_")),
               names_to = c(".value", "issue"),
               names_pattern = paste0("^(", k, "_alpha)_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
              names_prefix = "alpha_w", values_from = all_of(paste0(k, "_alpha")))

df_h15 <- df_h15_var |>
  left_join(df_h15_alpha, by = c("id", "issue")) |>
  left_join(df_p |> filter(wave == 2) |> select(id, treatment_wave2), by = "id")

m_h15 <- lm(alpha_w2 ~ alpha_w1 + rel_delta_var, data = df_h15)
cf_h15 <- summary(m_h15)$coefficients["rel_delta_var", ]

print_one_sided(
  sprintf("H15: Relative Delta social circle %s decreases issue weight (%s)", variance_type, k),
  cf_h15["Estimate"], cf_h15["Std. Error"], cf_h15["t value"],
  df = m_h15$df.residual,
  direction = "negative"
)
```

=== H15: Relative Delta social circle std decreases issue weight (exp) ===

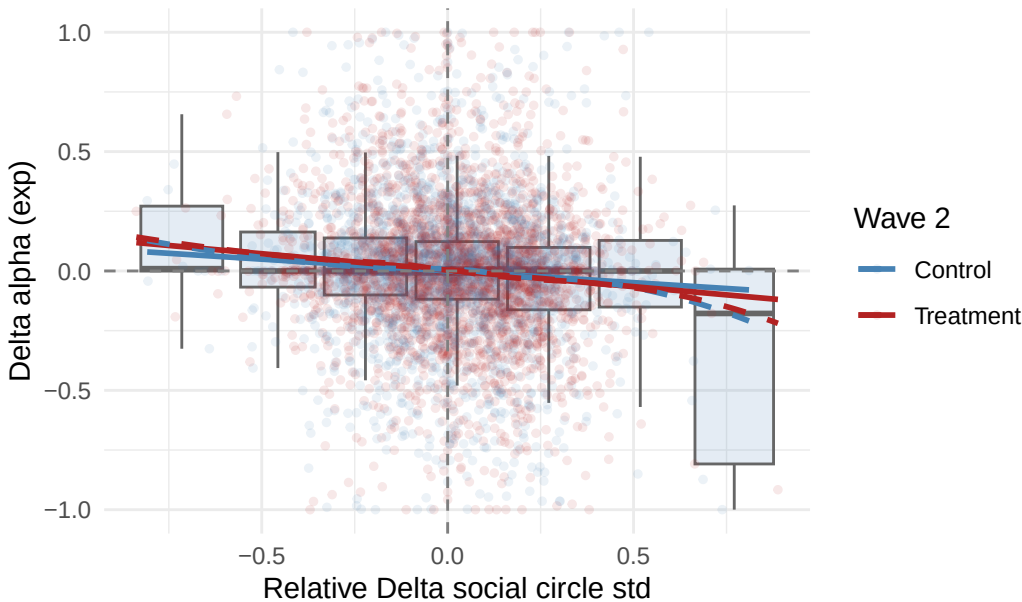
```
Estimate = -0.0595
SE       =  0.0143
t        = -4.1464
df       = 5373.0
p (one-sided, beta < 0) = 0.0000
Supported : YES
```



```
df_h15 |>
  filter(!is.na(rel_delta_var), !is.na(alpha_w2), !is.na(alpha_w1)) |>
  mutate(
    delta_alpha      = alpha_w2 - alpha_w1,
    rel_delta_var_bin = cut(rel_delta_var, breaks = 7),
    treatment_wave2   = factor(treatment_wave2, levels = c(0, 1),
                              labels = c("Control", "Treatment"))
  ) |>
  ggplot(aes(x = rel_delta_var, y = delta_alpha, color = treatment_wave2)) +
  geom_boxplot(aes(group = rel_delta_var_bin), fill = "steelblue", alpha = 0.15,
    outlier.shape = NA, color = "gray40") +
  geom_point(alpha = 0.1, size = 1) +
  geom_smooth(formula = y ~ x, method = "lm", se = FALSE, alpha = 0.5) +
  geom_smooth(formula = y ~ x, method = "loess", se = FALSE, linetype = "dashed", alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  scale_color_manual(values = c("Control" = "steelblue", "Treatment" = "firebrick"),
    name = "Wave 2") +
  labs(title = sprintf("H15: Relative social-circle-%s change vs weight (%s) change",
    variance_type, k),
    x = sprintf("Relative Delta social circle %s", variance_type),
    y = sprintf("Delta alpha (%s)", k)) +
  theme_minimal()
```

Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.

H15: Relative social-circle-std change vs weight (exp) chang



3.4.3 H16 (caution)

3.4.3.1 H16. Increase in overall social circle variance decreases distance to typical voters.

We model the map distances between the participant themselves and typical voters of political parties r in wave $w2$ as

$$d_{n, self-r}(w2) = d_{n, self-r}(w1) + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n, self-r, q} \cdot (\text{var}_{n, q}(w2) - \text{var}_{n, q}(w1)) + \epsilon_{n, r} \quad (25)$$

The explanatory variable is the change in social circle variance averaged over all six issues and weighted by the corresponding belief differences between participant and the typical voter of party r , $\Delta x_{n, self-r, q} = |x_{n, self, q} - x_{n, r, q}|$.

The prediction is $\beta_1 < 0$. When overall social circle variance increases, the reported distance between participant and typical voter beliefs decreases. We test this via a t-test with significance threshold $\alpha = 0.05$.

ERROR: This should not include the treatment. The effect we want to test is whether variance increase (weighted by opinion difference, such that) changes distances.

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \sum_q \Delta x_{n,self-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + \beta_2 \cdot \text{condition}_n + u_n + \epsilon_{n,r} \quad (26)$$

or alternatively, but that's not really a fair thing to test and not really related to our theory (see discussion with Mirta):

$$d_{n,self-r}(w2) = d_{n,self-r}(w1) + \beta_1 \cdot \text{condition}_n + \epsilon_{n,r} \quad (27)$$

```
# Build shared data objects used by both H16 and H17 variants

variance_type <- "std"
var_col       <- paste0(variance_type, "_socialCircle_ops")

# Wave 1 and 2 self-party distances
df_self_party_w1 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 1) |>
  select(id, dot2, pixel_dist_w1 = pixel_dist)

df_self_party_w2 <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, pixel_dist_w2 = pixel_dist, sympathy_w2 = sympathy)

# deltaX (wave 2)
df_deltaX <- df_diff |>
  filter(id %in% inds_bothwaves, dot1 == "self", dot2 %in% partiesVars, wave == 2) |>
  select(id, dot2, all_of(col_deltas))

# Variance change per participant per issue (wide)
df_delta_var_wide <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
    names_prefix = "var_w", values_from = all_of(var_col)) |>
  mutate(delta_var = var_w2 - var_w1) |>
  pivot_wider(id_cols = id, names_from = issue,
    names_prefix = "delta_var_", values_from = delta_var)

# Long delta_var (for new H16/H17 moderation models)
df_delta_var_long <- df_p |>
  filter(id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0(var_col, "_")),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(", var_col, ")_(.*)")) |>
  pivot_wider(id_cols = c(id, issue), names_from = wave,
    names_prefix = "var_w", values_from = all_of(var_col)) |>
  mutate(delta_var = var_w2 - var_w1,
    delta_var_c = as.numeric(scale(delta_var, center = TRUE, scale = FALSE))) |>
  select(id, issue, delta_var, delta_var_c)

# Simple direct-effect check (does treatment affect pixel_dist?)
df_h16_simple <- df_self_party_w2 |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control")) |>
  left_join(df_self_party_w1, by = c("id", "dot2"))

m_h16_direct <- lmer(pixel_dist_w2 ~ group + (1 | id) + (1 | dot2),
  data = df_h16_simple, REML = TRUE)
cat("=== H16 (simple: treatment -> pixel_dist) ===\n")
print(summary(m_h16_direct)$coefficients)

# Full model: weighted variance change
```

```
df_h16_full <- df_deltaX |>
  left_join(df_self_party_w1, by = c("id", "dot2")) |>
  left_join(df_self_party_w2, by = c("id", "dot2")) |>
  left_join(df_delta_var_wide, by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(col_deltas)) * across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    )
  )

m_h16_full <- lmer(pixel_dist_w2 ~ pixel_dist_w1 + weighted_var_change + group + (1 | id),
  data = df_h16_full, REML = TRUE)
cf_h16 <- lmer_coef(m_h16_full, "weighted_var_change")

cat("\n")
print_one_sided("H16 (full): weighted Deltavar decreases perceived distance",
  cf_h16$beta, cf_h16$se, cf_h16$t, direction = "negative")
```

```
=== H16 (simple: treatment → pixel_dist) ===
              Estimate Std. Error      df    t value      Pr(>|t|)
(Intercept)   0.338015954 0.017448847   7.643035 19.3718221 8.988148e-08
grouptreatment -0.006245272 0.008341309 900.000072 -0.7487161 4.542240e-01
```

```
=== H16 (full): weighted Deltavar decreases perceived distance ===
Estimate = 0.0082
SE        = 0.0046
t         = 1.7734
df        = infty
p (one-sided, beta < 0) = 0.9619
Supported : NO
```

3.4.4 H17 (caution)

3.4.4.1 H17. Increase in overall social circle variance increases likeability rating of typical voters.

Similarly, we model how participant n rates the likeability rating of typical voters of party r in wave 2 as

$$y_{n,r} = \beta_0 + \beta_1 \cdot \mathbb{I}(\text{condition}_n = T) \cdot \sum_q \Delta x_{n,\text{self}-r,q} \cdot (\text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)) + u_n + \epsilon_{n,r} \quad (28)$$

The prediction is $\beta_1 > 0$. When overall social circle variance increases, the likeability rating of typical voters of the political parties increases. %We test this via a t-test with significance threshold $\alpha = 0.05$.

Most simple: Does treatment affect liking? No obviously not!

```
# Simple direct-effect check
df_h17_simple <- df_self_party_w2 |>
  mutate(group = if_else(id %in% treatment_ids, "treatment", "control"))

m_h17_direct <- lmer(sympathy_w2 ~ group + (1 | id) + (1 | dot2),
  data = df_h17_simple, REML = TRUE)
cat("=== H17 (simple: treatment → sympathy) ===\n")
print(summary(m_h17_direct)$coefficients)

# Full model
df_h17_full <- df_deltaX |>
  left_join(df_self_party_w2, by = c("id", "dot2")) |>
  left_join(df_delta_var_wide, by = "id") |>
  mutate(
    group = if_else(id %in% treatment_ids, "treatment", "control"),
    weighted_var_change = rowSums(
      across(all_of(col_deltas)) * across(all_of(paste0("delta_var_", questions_sc))),
      na.rm = TRUE
    )
  )
```

```

)

m_h17_full <- lmer(sympathy_w2 ~ weighted_var_change + group + (1 | id),
  data = df_h17_full, REML = TRUE)
cf_h17 <- lmer_coef(m_h17_full, "weighted_var_change")

cat("\n")
print_one_sided("H17 (full): weighted Deltavar increases likeability",
  cf_h17$beta, cf_h17$se, cf_h17$t, direction = "positive")

=== H17 (simple: treatment → sympathy) ===
      Estimate Std. Error      df    t value      Pr(>|t|)
(Intercept)   0.423891501 0.026777532   6.507399 15.8301182 1.939544e-06
grouptreatment -0.003302603 0.008775184 899.999845 -0.3763571 7.067402e-01

=== H17 (full): weighted Deltavar increases likeability ===
Estimate = -0.0302
SE        = 0.0075
t         = -4.0329
df        = infty
p (one-sided, beta > 0) = 1.0000
Supported : NO

```

3.5 NEW: new versions of H16 and H17 (moderation)

3.5.1 H16 new

3.5.1.1 H16: Treatment increases social circle variance, which weakens the effect of opinion differences on perceived distance.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q} = |x_{n,\text{self},q} - x_{n,r,q}|$ and perceived distance d ? We model:

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (29)$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 < 0$: the treatment weakens the effect of opinion differences on perceived distance.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta \text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \rightarrow d$ relationship, above and beyond treatment assignment?

$$d_{n,\text{self}-r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta \text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta \text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (30)$$

The prediction is $\beta_3 < 0$: when social circle variance increases, the effect of opinion differences on perceived distance weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta \text{var} \xrightarrow{\beta_3 < 0} \text{weakens } (\Delta X \rightarrow d) \quad (31)$$

```

# Step 1 - ITT: does treatment moderate DeltaX → distance?
df_h16_itt <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2,
    id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0("^(\deltaX|", var_col, ")_((", paste(questions_sc, collapse = "|"), ")$"))),
    names_to = c(".value", "issue"),
    names_pattern = paste0("^(\deltaX|", var_col, ")_(.*)")) |>
  select(id, wave, dot2, pixel_dist, issue, deltaX, all_of(var_col)) |>
  mutate(treatment = as.integer(id %in% treatment_ids))

m_h16_itt <- lmer(pixel_dist ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 | dot2),
  data = df_h16_itt)
cf_itt_16 <- lmer_coef(m_h16_itt, "deltaX:treatment")

```

```

cat("\n")
print_one_sided("H16-new Step 1 (ITT): treatment weakens DeltaX → distance",
               cf_itt_16$beta, cf_itt_16$se, cf_itt_16$t, direction = "negative")

# Step 2 - Deltavar as moderator
df_h16_var <- df_h16_itt |>
  left_join(df_delta_var_long, by = c("id", "issue"))

m_h16_var <- lmer(pixel_dist ~ deltaX * delta_var_c + (1 | id) + (1 | issue) + (1 | dot2),
                 data = df_h16_var)
cf_var_16 <- lmer_coef(m_h16_var, "deltaX:delta_var_c")

print_one_sided("H16-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → distance",
               cf_var_16$beta, cf_var_16$se, cf_var_16$t, direction = "negative")

```

```

=== H16-new Step 1 (ITT): treatment weakens DeltaX → distance ===

```

```

Estimate = 0.0044
SE        = 0.0030
t         = 1.4548
df        = infty
p (one-sided, beta < 0) = 0.9271
Supported : NO

```

```

=== H16-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → distance ===

```

```

Estimate = 0.0035
SE        = 0.0071
t         = 0.4890
df        = infty
p (one-sided, beta < 0) = 0.6876
Supported : NO

```

3.5.2 H17 new

3.5.2.1 H17: Treatment increases social circle variance, which increases likeability ratings of typical party voters.

We test this hypothesis in two steps, conditional on H14 being confirmed (i.e., the nudge successfully increased social circle variance in the treatment group).

Step 1 — Intent-to-Treat (ITT): Does the treatment moderate the relationship between opinion differences $\Delta X_{n,r,q}$ and likeability ratings $y_{n,r}$? We model:

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \mathbb{I}(\text{condition}_n = T) + \beta_3 \cdot \Delta X_{n,r,q} \cdot \mathbb{I}(\text{condition}_n = T) + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (32)$$

where u_n is the participant random intercept, v_r the party voter random intercept, and v_q the question random intercept. The prediction is $\beta_3 > 0$: the treatment weakens the negative effect of opinion differences on likeability, i.e. participants in the treatment condition rate typical voters of parties they disagree with more favourably.

Step 2 — Variance as moderator: Does the actual change in variance $\Delta \text{var}_{n,q} = \text{var}_{n,q}(w2) - \text{var}_{n,q}(w1)$ moderate the $\Delta X \rightarrow y$ relationship, above and beyond treatment assignment?

$$y_{n,r,q}(w2) = \beta_0 + \beta_1 \cdot \Delta X_{n,r,q} + \beta_2 \cdot \Delta \text{var}_{n,q} + \beta_3 \cdot \Delta X_{n,r,q} \cdot \Delta \text{var}_{n,q} + u_n + v_r + v_q + \epsilon_{n,r,q} \quad (33)$$

The prediction is $\beta_3 > 0$: when social circle variance increases, the negative effect of opinion differences on likeability weakens. Step 2 speaks to the *mechanism*: it is the variance increase itself, not merely treatment assignment, that drives the moderation. Together, Steps 1 and 2 establish the causal chain:

$$\mathbb{I}(T) \xrightarrow{H14} \Delta \text{var} \xrightarrow{\beta_3 > 0} \text{weakens } (\Delta X \rightarrow -y) \quad (34)$$

```

df_h17_itt <- df_diff |>
  filter(dot1 == "self", dot2 %in% partiesVars, wave == 2, # fixed: was "2"
         id %in% c(treatment_ids, control_ids)) |>
  pivot_longer(cols = matches(paste0("^(\deltaX|", var_col, ")_(", paste(questions_sc, collapse = "|"), ")$"))),
               names_to = c(".value", "issue"),
               names_pattern = paste0("^(\deltaX|", var_col, ")_(.*)")) |>

```

```

select(id, wave, dot2, sympathy, issue, deltaX, all_of(var_col)) |>
mutate(treatment = as.integer(id %in% treatment_ids))

m_h17_itt <- lmer(sympathy ~ deltaX * treatment + (1 | id) + (1 | issue) + (1 | dot2),
  data = df_h17_itt)
cf_itt_17 <- lmer_coef(m_h17_itt, "deltaX:treatment")

cat("\n")
print_one_sided("H17-new Step 1 (ITT): treatment weakens DeltaX → low likeability",
  cf_itt_17$beta, cf_itt_17$se, cf_itt_17$t, direction = "positive")

df_h17_var <- df_h17_itt |>
  left_join(df_delta_var_long, by = c("id", "issue"))

m_h17_var <- lmer(sympathy ~ deltaX * delta_var_c + (1 | id) + (1 | issue) + (1 | dot2),
  data = df_h17_var)
cf_var_17 <- lmer_coef(m_h17_var, "deltaX:delta_var_c")

print_one_sided("H17-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → low likeability",
  cf_var_17$beta, cf_var_17$se, cf_var_17$t, direction = "positive")

```

```

=== H17-new Step 1 (ITT): treatment weakens DeltaX → low likeability ===

```

```

Estimate = -0.0168
SE        =  0.0053
t         = -3.1633
df        = infty
p (one-sided, beta > 0) = 0.9992
Supported : NO

```

```

=== H17-new Step 2 (Deltavar moderator): Deltavar weakens DeltaX → low likeability ===

```

```

Estimate = -0.0256
SE        =  0.0126
t         = -2.0382
df        = infty
p (one-sided, beta > 0) = 0.9792
Supported : NO

```

```

#

```

4 Exclusion criteria

Participants are excluded if they complete the survey in less than one third of the median completion time, provide invariant slider responses, or do not indicate their own beliefs on the six political issues.